Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

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Supplementary Content

S1 Proof of Theorem 1 (DMOP-α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x^{A,P,\theta}_{1:n,j}$ contains the history of the prediction particle $x^{P,\theta}_{n,j}$ at observation times $1:n$. Similarly, $g^{A,P,\theta}_{i,j}$ is the ancestral weight of $x^{P,\theta}_{n,j}$ at observation time i.

Lemma S1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,F,\theta}_{1:n,j};\theta\big), \tag{S1}$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider the case of MOP-α when $\alpha = 1$ and $\theta = \phi$. We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned}\hat{\mathcal{L}}^1(\theta) \quad &:= \quad \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi}\cdot\frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} = \prod_{n=1}^{N} L_n^{\phi}\cdot\frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{F,\theta}_{n-1,j}} \\ &= \quad \left(\frac{1}{J}\sum_{j=1}^{J} w^{F,\theta}_{N,j}\right)\prod_{n=1}^{N} L_n^{\phi},\end{aligned} \tag{S2}$$

where the third equality follows from the choice of $\alpha = 1$, and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_\theta\left(\frac{1}{J}\sum_{j=1}^J w^{F,\theta}_{N,j}\right)\prod_{n=1}^N L_n^\phi}{\prod_{n=1}^N L_n^\phi} = \frac{1}{J}\sum_{j=1}^{J}\nabla_\theta w^{F,\theta}_{N,j}. \tag{S3}$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_\theta \hat{\ell}^1(\theta) := \frac{1}{J}\sum_{j=1}^{J}\nabla_\theta w^{F,\theta}_{N,j}. \tag{S4}$$

We proceed to decompose (S4). First, observe that as $\alpha = 1$,

$$w^{P,\theta}_{n,j} = w^{F,\theta}_{n-1,j}\frac{g^{\theta}_{n,j}}{g^{\phi}_{n,j}} = \prod_{i=1}^{n}\frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}, \tag{S5}$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the j-th prediction particle at timestep n. We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_\theta w^{P,\theta}_{n,j}}{w^{P,\theta}_{n,j}} = \nabla_\theta \log w^{P,\theta}_{n,j} \quad = \quad \nabla_\theta \log\left(\prod_{i=1}^{n}\frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}\right) \tag{S6}$$

$$= \nabla_\theta \sum_{i=1}^{n} \left(\log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi}\right) \quad \text{(S7)}$$

$$= \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,P,\theta}. \quad \text{(S8)}$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the j-th prediction particle up to timestep n,

$$\nabla_\theta \sum_{n=1}^{N} \log g_{n,j}^{A,P,\theta} = \nabla_\theta \log \left(\prod_{n=1}^{N} g_{n,j}^{A,P,\theta}\right) \quad \text{(S9)}$$

$$= \nabla_\theta \log \left(\prod_{n=1}^{N} f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{A,P,\theta}; \theta\big)\right) \quad \text{(S10)}$$

$$= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta\big). \quad \text{(S11)}$$

Multiplying both sides of the expression by $w_{N,j}^{P,\theta}$ yields an expression for the gradient of the weights at timestep N,

$$\nabla_\theta w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta\big). \quad \text{(S12)}$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J} \sum_{j=1}^{J} w_{N,k_j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta\big). \quad \text{(S13)}$$

Finally, observing that $\theta = \phi$ implies $w_{N,j}^{F,\theta} = 1$, we obtain

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta\big). \quad \text{(S14)}$$

This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. □

Note that the variance of the DMOP-α log-likelihood derivative estimate scales poorly with N when $\theta \neq \phi$. This can be seen by observing that

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta}$$

$$= \quad \frac{1}{J}\sum_{j=1}^{J} w^{P,\theta}_{N,k_j}\nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,P,\theta}_{1:n,k_j};\theta\big). \tag{S15}$$

When $\theta \neq \phi$, we see that $w^{P,\theta}_{N,k_j} = O(c^N)$. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite-J upper bound.

Lemma S2. *Write $\nabla_\theta\hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta\hat{\ell}^0(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\Big(f_{Y_n|X_n}\big(y^*_n|x^{F,\theta}_{n,j};\theta\big)\Big), \tag{S16}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)} \tag{S17}$$

as shorthand for the measurement density ratios. Observe that, when $\alpha = 0$,, the likelihood estimate becomes

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} \quad = \quad \prod_{n=1}^{N} L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} \tag{S18}$$

$$= \quad \prod_{n=1}^{N} L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^{J} s_{n,j} \tag{S19}$$

$$= \quad \prod_{n=1}^{N} L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^{J} \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)}. \tag{S20}$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta\hat{\ell}^0(\theta) := \sum_{n=1}^{N}\nabla_\theta \log\left(L_n^\phi \frac{1}{J}\sum_{j=1}^{J} s_{n,j}\right) \tag{S21}$$

$$= \sum_{n=1}^{N} \frac{\nabla_\theta\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)}{\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)} \tag{S22}$$

$$= \sum_{n=1}^{N} \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \tag{S23}$$

$$= \sum_{n=1}^{N} \frac{1}{J} \sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\theta}; \theta\big)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \tag{S24}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \Big(f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\theta}; \theta\big) \Big), \tag{S25}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. $\square$

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n, the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^\theta(Z_n) = f_{Y_n|X_n}(y_n^* \mid X_n^\theta; \theta), \qquad W_\theta(Z_N) = \prod_{n=1}^{N} G_n^\theta(Z_n), \tag{S26}$$

$$\psi_\theta(Z_N) = \nabla_\theta^{\text{tot}} \log W_\theta(Z_N), \qquad \Xi_\theta(Z_N) = \nabla_\theta^{2,\text{tot}} \log W_\theta(Z_N). \tag{S27}$$

The superscript "tot" means that the derivative passes through the reparameterized simulated path while the base random numbers and the discrete resampling indices are held fixed.

Write $\rho_N^J = J^{-1} \sum_{j=1}^{J} \delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_\theta(Z_N) F(Z_N)\}}{\mathbb{E}\{W_\theta(Z_N)\}}. \tag{S28}$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_\theta| \le N \bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_\theta \ell(\theta) = \rho_N(\psi_\theta), \qquad \nabla_\theta \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_\theta). \tag{S29}$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1, \ldots, N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \le q \le N$, write $\rho_q^J = J^{-1} \sum_{j=1}^{J} \delta_{\mathsf{Z}_{q,j}}$, where the $\mathsf{Z}_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $\mathsf{Z}_{N,j} = Z_{N,j}$. Write $V_q^\theta(\mathsf{Z}_q) = G_{q+1}^\theta(\mathsf{Z}_q)$ for $q = 0, \ldots, N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^\theta$ append the next independent simulator innovation and its gradients for $0 \le q \le N-2$, and put $\mathcal{M}_N^\theta = \mathrm{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^\theta(\mu)(dz') = \frac{\int \mu(dz) V_q^\theta(z) \mathcal{M}_{q+1}^\theta(z, dz')}{\mu(V_q^\theta)}, \qquad 0 \le q < N. \tag{S30}$$

Conditionally on the current cloud, the J children are independent with common law $\Phi^\theta_{q+1}(\rho^J_q)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V^\theta_{N-1} = G^\theta_N$ followed by the identity mutation. Thus ρ^J_N is exactly the terminal empirical measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \le q \le N$, let

$$Q^\theta_{q,N}F(z_q) = \mathbb{E}\left[F(\mathsf{Z}_N)\prod_{r=q}^{N-1} V^\theta_r(\mathsf{Z}_r) \,\middle|\, \mathsf{Z}_q = z_q\right], \tag{S31}$$

$$P^\theta_{q,N}F = \frac{Q^\theta_{q,N}F}{Q^\theta_{q,N}1}, \qquad g_{q,N} = \sup_{z,z'} \frac{Q^\theta_{q,N}1(z)}{Q^\theta_{q,N}1(z')}, \qquad b_{q,N} = \sup_{\operatorname{osc}(F)\le 1} \operatorname{osc}(P^\theta_{q,N}F), \tag{S32}$$

where $Q^\theta_{N,N}F = F$ and $\operatorname{osc}(F) = \sup_{z,z'}|F(z) - F(z')|$. Thus $0 \le b_{q,N} \le 1$. With

$$R_G = \frac{\bar{G}}{\underline{\mathrm{G}}}, \qquad S_N = \sum_{q=0}^{N} g^3_{q,N} b_{q,N}, \qquad K_N = 32 S_N, \qquad K^{\mathrm{pf}}_N = 32\sum_{r=0}^{N} R^{3r}_G, \tag{S33}$$

the potential bounds give, term by term,

$$\underline{\mathrm{G}}^{N-q} \le Q^\theta_{q,N}1(z_q) \le \bar{G}^{N-q}, \qquad g_{q,N} \le R^{N-q}_G, \qquad K_N \le K^{\mathrm{pf}}_N. \tag{S34}$$

In particular, the constant used in the theorem is the closed-form quantity

$$K^{\mathrm{pf}}_N = \begin{cases} 32(N+1), & R_G = 1, \\ 32\dfrac{R^{3(N+1)}_G - 1}{R^3_G - 1}, & R_G > 1. \end{cases} \tag{S35}$$

Here is the constant calculation behind the tail bound used below. Let r^{DM}_N and β^{DM}_N denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r^{\mathrm{DM}}_N \le 4S_N, \qquad (\beta^{\mathrm{DM}}_N)^2 \le 4\sum_{q=0}^{N} g^2_{q,N} b^2_{q,N} \le 4S_N, \tag{S36}$$

where the last inequality uses $g_{q,N} \ge 1$ and $0 \le b_{q,N} \le 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \le \log 6$, then $6e^{-x} \ge 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \le 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \le 8x$ to obtain

$$\frac{r^{\mathrm{DM}}_N}{J}\{1 + h^{-1}(x)\} \le \frac{u^2}{8x}\{1 + 2x + 2\sqrt{x}\} \le u^2 \le \frac{u}{2},$$
$$\beta^{\mathrm{DM}}_N\sqrt{\frac{2x}{J}} \le 2\sqrt{S_N}\sqrt{\frac{2x}{J}} = \frac{u}{2}.$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr\left(|(\rho_N^J - \rho_N)(F)| > \operatorname{osc}(F)u\right) \le 6\exp\left(-\frac{Ju^2}{K_N}\right) \le 6\exp\left(-\frac{Ju^2}{K_N^{\mathrm{pf}}}\right). \tag{S37}$$

for every F with $\operatorname{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho_N^J - \rho_N)(F) = 0$ identically. Equations S33– S37 replace all unnamed model-dependent concentration constants.

Lemma S3 (Explicit concentration of the DMOP-1 gradient)**.** *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1\le i\le p} \operatorname{osc}(\psi_{\theta,i}), \qquad u_g(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6p}{\delta}}. \tag{S38}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta \hat\ell_J^{A,1}(\theta) - \nabla_\theta \ell(\theta)\|_2 \le A_N(\theta)\sqrt{p}\,u_g(J,\delta). \tag{S39}$$

Moreover, Assumption (A3) gives $A_N(\theta) \le 2NG'(\theta)$. Consequently, the explicit condition

$$J \ge K_N^{\mathrm{pf}}\log\frac{6p}{\delta}\max\left\{4, \frac{4pN^2G'(\theta)^2}{\epsilon^2}\right\} \tag{S40}$$

implies that the left side of equation S39 is at most ϵ with probability at least $1-\delta$.

Proof. The lower bound on J makes $u_g(J,\delta) \le 1/2$. Apply equation S37 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1\le i\le p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \le A_N(\theta)u_g(J,\delta)$$

with probability at least $1-\delta$. Now use $\|v\|_2 \le \sqrt{p}\|v\|_\infty$ and equation S29. Finally, $|\psi_{\theta,i}| \le NG'(\theta)$, so $A_N(\theta) \le 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}\,u_g(J,\delta) \le \epsilon$ gives equation S40. $\square$

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian)**.** *Suppose in addition that Assumption (A3) gives the entrywise bound $|\Xi_{\theta,ab}(Z_N)| \le NG''(\theta)$ for $1 \le a,b \le p$. Define*

$$d_p = \frac{p(p+3)}{2}, \qquad H_N(\theta) = 2p\{NG''(\theta) + 3N^2G'(\theta)^2\}, \qquad u_H(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6d_p}{\delta}}. \tag{S41}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6d_p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta^2 \hat\ell_J^{A,1}(\theta) - \nabla_\theta^2 \ell(\theta)\|_{\mathrm{op}} \le H_N(\theta)u_H(J,\delta). \tag{S42}$$

Fix $0 < \kappa < \gamma$. *If*

$$\gamma I_p \preceq -\nabla^2_\theta \ell(\theta) \preceq \Gamma I_p \quad \textit{and} \quad J \ge K_N^{\mathrm{pf}} \log \frac{6d_p}{\delta} \max\left\{4, \frac{H_N(\theta)^2}{\kappa^2}\right\}, \tag{S43}$$

then, with probability at least $1-\delta$,

$$(\gamma-\kappa) I_p \preceq -\nabla^2_\theta \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma+\kappa) I_p. \tag{S44}$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log\left\{\frac{1}{J}\sum_{j=1}^J \exp(a_j(\theta))\right\}, \qquad a_j(\theta) = \log w_{N,j}^{F,\theta}, \qquad a_j(\phi) = 0,$$

where C_ϕ is constant in θ. Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla^2_\theta \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N^J(\psi_\theta)\rho_N^J(\psi_\theta)^T. \tag{S45}$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial^2_{\theta_a\theta_b} W_\theta| = W_\theta |\Xi_{\theta,ab} + \psi_{\theta,a}\psi_{\theta,b}| \le \bar{G}^N\{NG''(\theta) + N^2 G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E} W_\theta$ gives the population identity

$$\nabla^2_\theta \ell = \rho_N(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N(\psi_\theta)\rho_N(\psi_\theta)^T. \tag{S46}$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1, and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2. Apply equation S37 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned} \|(\rho_N^J - \rho_N)(\psi)\|_2 &\le A_1\sqrt{p}\, u_H(J,\delta), \\ \|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\mathrm{op}} &\le pA_2 u_H(J,\delta). \end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p}\, NG'(\theta)$. Therefore

$$\begin{aligned} &\|\rho_N^J(\psi)\rho_N^J(\psi)^T - \rho_N(\psi)\rho_N(\psi)^T\|_{\mathrm{op}} \\ &\qquad \le 2\sqrt{p}\, NG'(\theta)\, \|(\rho_N^J - \rho_N)(\psi)\|_2 \le 4pN^2G'(\theta)^2 u_H(J,\delta). \end{aligned}$$

Combining the last three displays with equation S45 and equation S46 proves equation S42. Condition equation S43 makes its right side at most κ; Weyl's eigenvalue inequality then gives equation S44. □

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\widetilde{B} = Q\,\mathrm{diag}(\lambda_1, \dots, \lambda_p)Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q\,\mathrm{diag}([\lambda_1]_c^C, \dots, [\lambda_p]_c^C)Q^T, \qquad [x]_c^C = \min\{C, \max\{c, x\}\}. \tag{S47}$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \le v^T B v \le C\|v\|_2^2, \qquad C^{-1}\|v\|_2^2 \le v^T B^{-1} v \le c^{-1}\|v\|_2^2. \tag{S48}$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S43; equation S47 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let τ and $\mathcal{E}_M$ be the stopping time and simultaneous gradient event defined in Theorem 2. If $\mathcal{F}_m$ is the history before the mth particle-filter call, then θ_m and $\{m \le \min(\tau, M-1)\}$ are $\mathcal{F}_m$-measurable. Lemma S3, with failure probability δ/M, gives

$$\begin{aligned}\Pr(\mathcal{E}_M^c \mid \theta_0) &\le \sum_{m=0}^{M-1} \mathbb{E}\left[\mathbf{1}_{\{m \le \min(\tau, M-1)\}} \Pr\left(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m\right) \big| \, \theta_0\right] \\ &\le \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta.\end{aligned} \tag{S49}$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1} g_m$. If $\|g_m\|_2 \le \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \le \|g_m\|_2 + \epsilon \le (\sigma + 1)\epsilon. \tag{S50}$$

Strong convexity implies $f(\theta) - f(\theta^*) \le \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \le \frac{(\sigma+1)^2\epsilon^2}{2\gamma}. \tag{S51}$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \le \|g_m\|_2 \le C\|p_m\|_2, \qquad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \tag{S52}$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned}f(\theta_m + \eta p_m) - f(\theta_m) &\le \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2 \\ &\le -\eta p_m^T B_m p_m + \eta\epsilon\|p_m\|_2 + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2.\end{aligned} \tag{S53}$$

The choices of σ and η imply

$$\frac{C}{\sigma}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2, \qquad \frac{\Gamma\eta}{2}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c\|p_m\|_2^2$ in equation S53 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \tag{S54}$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma+1}{\sigma}\|g_m\|_2. \tag{S55}$$

Consequently, equation S48 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \tag{S56}$$

Combining equation S54 and equation S56 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q\{f(\theta_m) - f(\theta^*)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2. \tag{S57}$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1-\beta)\frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \qquad \text{so} \qquad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S51 gives its stopping conclusion after substituting $f = -\ell$. □

The result is deliberately restricted to DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$ be sequences of probability measures on the state space $\mathcal{X}$. This is the sequence of prediction distributions $f_{X_n|Y_{1:n-1}}$, filtering distributions $f_{X_n|Y_{1:n}}$, and posterior distributions $f_{X_{1:n}|Y_{1:n}}$ that we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0,1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h \, d\eta = \int h(x)\eta(dx), \tag{S58}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{S59}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{S60}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \tag{S61}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \le 1/2} |\mu(\phi) - \nu(\phi)| = \sup_{\mathrm{osc}(h)\le 1} |\mu(h) - \nu(h)|. \tag{S62}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y\in\mathcal{X}} \big\|M(x, \cdot) - M(y, \cdot)\big\|_{\mathrm{TV}} = \sup_{\mu,\nu\in\mathcal{P},\mu\neq\nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{S63}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y^*_n|x^F_{n,j};\theta\big) = G_n(x^F_{n,j})$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y^*_n|\,\cdot\,)$ is the conditional density of the observed measurement at time n, where we condition on the filtering particle $x^F_{n,j}$ as an element of the state space.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{S64}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{S65}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ ... \circ \Phi_{k+1}. \tag{S66}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^P}, \tag{S67}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^F}. \tag{S68}$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \left(\Phi_n\left(\eta_{n-1}^J\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J} \delta_{x_{n-1,j}^P}\right)\right)^{\otimes J}, \tag{S69}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \tag{S70}$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on $\mathcal{X}^J$. This is because at every timestep n, we in fact draw $X_{n,j}^P | \{X_{n-1,1:J}^P = x_{n-1,1:J}^P\} \sim \mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \eta_0^{\otimes J} \mathbf{M}_{0,n}$ for $j = 1, ..., J$.

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP-α. That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \tag{S71}$$

for some ϵ dependent on $\bar{G}, \underline{G}, \bar{M}, \underline{M}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP-α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP-α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters). *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model* $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$*, where we inductively define given some* $x_{n-1,j}^F$, $w_{n-1,j}^F$ *comprising the filtering measure approximation* π_n^J*:*

1. *The prediction particles at timestep* n, $x_{n,j}^P$, *are drawn from* $X_{n,j}^P \sim \pi_n^J M_n$, *and the prediction weights are given by* $w_{n,j}^P = w_{n-1,j}^F$.
2. *The prediction measure at timestep* n *is given by*
$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \, \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \tag{S72}$$
3. *The particles are resampled at every timestep* n, *yielding indices* k_j *according to some arbitrary probabilities* $(p_{n,j})_{j=1}^J$.
4. *The filtering measure at timestep* n *is given by*
$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P \, g_{n,j}}{\sum_{j=1}^J w_{n,j}^P \, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \, \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \tag{S73}$$
5. *The prediction weights at the next timestep are given by* $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}$.

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting). *A pair of random vectors* (X, W) *drawn from some measure* g ***targets*** *another measure* π *if, for any measurable and bounded functional* h,
$$E_g\big[h(X) \cdot W\big] = E_\pi\big[h(X)\big]. \tag{S74}$$
A particle approximation is a triangular array of pairs of random vectors (X_j^J, W_j^J), $j = 1, 2, \ldots, J$. *The particle approximation* ***targets*** π *if, for any measurable and bounded functional* h,
$$\frac{\sum_{j=1}^J h(X_j^J)\, W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{a.s.} E_\pi\big[h(X)\big] \tag{S75}$$
as $J \to \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure)**.** *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ targets f_X. Now, let $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ be a multinomial sample with indices k_j drawn from $\{(\tilde{X}_j^J, U_j^J)\}$ where $(\tilde{X}_j^J, U_j^J)$ is represented, on average, proportional to $\pi_j^J J$ times. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{S76}$$

If the importance sampling weights U_j/π_j are bounded, then $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ targets f_X.

Proof. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{S77}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f(X)\big]. \tag{S78}$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{S79}$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \pi_j^J \overset{a.s.}{\to} 0 \tag{S80}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S79). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \Big(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J)\, \pi_j^J \Big) \overset{a.s.}{\to} 0. \tag{S81}$$

Because the functional h and importance sampling weights $u^J_{k_j}/\pi^J_{k_j}$ are bounded, we have that $\mathbb{E}\big[(Z^J_j)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z^J_j are conditionally independent given the $\tilde{X}^J_j$ and U^J_j,

$$\begin{aligned}\mathbb{E}\left[\left(\sum_j Z^J_j\right)^4 \middle| (\tilde{X}^J_j, U^J_j)^J_{j=1}\right] &= J\,\mathbb{E}\left[(Z^J_1)^4 | (\tilde{X}^J_j, U^J_j)^J_{j=1}\right] \\ &\quad + 3J(J-1)\left(\mathbb{E}\big[(Z^J_j)^2 \big| (\tilde{X}^J_j, U^J_j)^J_{j=1}\big]\right) && \text{(S82)} \\ &\leq CJ^2, && \text{(S83)}\end{aligned}$$

for some $C > 0$. Taking expectations on both sides yields

$$\mathbb{E}\left[\left(\sum_j Z^J_j\right)^4\right] \leq CJ^2 \tag{S84}$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P}\left(\left|\frac{1}{J}\sum_{j=1}^J Z^J_j\right| > \epsilon\right) \leq \frac{1}{\epsilon^4 J^4}\mathbb{E}\left[\left(\sum_{j=1}^J Z^J_j\right)^4\right] \leq \frac{C}{\epsilon^4 J^2}, \tag{S85}$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\sum_j h(X^J_{k_j})\,\frac{u^J_{k_j}}{\pi^J_{k_j}} - \sum_j h(X^J_j)\,\frac{u^J_j}{\pi^J_j}\,\pi^J_j \xrightarrow{a.s.} 0. \tag{S86}$$

Similarly, we also have that

$$\sum_j \frac{u^J_j}{\pi^J_j}\pi^J_j - \sum_j \frac{u^J_{k_j}}{\pi^J_{k_j}} \xrightarrow{a.s.} 0, \tag{S87}$$

and the result is proved.

□

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}^J_j, U^J_j), j = 1, \ldots, J\}$ targets f_X. Also suppose that $\tilde{Z}^J_j \sim f_{Z|X}(\cdot|\tilde{X}^J_j)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U^J_j are bounded, $\{(\tilde{Z}^J_j, U^J_j)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$\frac{\sum_j h(\tilde{Z}^J_j)\,U^J_j}{\sum_j U^J_j} \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\left[\mathbb{E}_{f_{Z|X}}[h(Z)|X]\right]. \tag{S88}$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{S89}$$

Let $\bar{U}_j^J = \frac{J\, U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} = \frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J}{J^{-1}\sum_j \bar{U}_j^J}. \tag{S90}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J, \tag{S91}$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\begin{aligned} \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J\right] &= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(\tilde{Z}_j^J)\, \bar{U}_j^J \Big| \tilde{X}_j^J, \bar{U}_j^J\Big]\right] && \text{(S92)} \\ &= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(Z)\Big| X = \tilde{X}_j^J\Big]\, \bar{U}_j^J\right]. && \text{(S93)} \end{aligned}$$

Applying (S89) with $g(x) = \mathbb{E}\big[h(Z)|X = x\big]$, the average on the right hand side converges almost surely to $\mathbb{E}\big\{\mathbb{E}[h(Z)|X]\big\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{S94}$$

□

Lemma S7 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$*. Then, if* $U_j^J\, f_{Z|X}(z^*|X_j^J)$ *and* $U_j^J\, f_{Z|X}(z^*|X_j^J)\big/f_Z(z^*)$ *are bounded,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$*.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J)\cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \overset{a.s.}{\to} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{S95}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{S96}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{S97}$$

We will apply Equation (S96) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j u_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big], \tag{S98}$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{S99}$$

Now we take advantage of the identities

$$\begin{aligned}
\frac{\mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} &= \mathbb{E}_{f_X}\left[\frac{h(X)\, f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] &\text{(S100)}\\
&= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\big[h(X)|z^*\big], &\text{(S101)}
\end{aligned}$$

to give the desired result.

□

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^F, w_{n-1,j}^F)\}$ targets π_{n-1}. Then, Lemma S6 tells us that $\{(X_{n,j}^P, w_{n,j}^P)\}$ targets η_n. Lemma S7 tells us that $\{(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta)\}$ therefore targets π_n. Lemma S5 guarantees that the resampling rule, given by

$$\big(X_{n,j}^F, w_{n,j}^F\big) = \big(X_{n,k_j}^P, w_{n,k_j}^P\, g_{n,k_j}/p_{n,k_j}\big), \tag{S102}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition S1 (MOP-1 Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, MOP-α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*;\theta)$. Then, Lemma S6 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. Lemma S7 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j}\, g^{\theta}_{n,j})\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. Lemma S5 guarantees that the resampling rule, given by

$$\big(X^{F,\theta}_{n,j}, w^{F,\theta}_{n,j}\big) = \big(X^{P,\theta}_{n,k_j}, w^{P,\theta}_{n,k_j}\, g^{\theta}_{n,k_j}/g^{\phi}_{n,k_j}\big), \tag{S103}$$

with resampling weights proportional to $g^{\phi}_{n,j}$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting). $f_{Y_n|Y_{1:n-1}}(y^*_n|y^*_{1_n-1};\theta)$ *is strongly consistently estimated by either the before-resampling estimate,*

$$L^{B,\theta}_n = \frac{\sum_{j=1}^J g^{\theta}_{n,j}\, w^{P,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}, \tag{S104}$$

or by the after-resampling estimate,

$$L^{A,\theta}_n = L^{\phi}_n \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}. \tag{S105}$$

where L^{ϕ}_n is as defined in the various algorithms.

Here, (S104) is a direct consequence of our earlier result that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. To see (S105), we write the numerator of (S5) as

$$L^{\phi}_n \sum_{j=1}^J \left[\frac{g^{\theta}_{n,j}}{g^{\phi}_{n,j}}\, w^{P,\theta}_{n,j}\right] \frac{g^{\phi}_{n,j}}{L^{\phi}_n} = L^{\phi}_n \sum_{j=1}^J w^{FC,\theta}_{n,j}\, \frac{g^{\phi}_{n,j}}{L^{\phi}_n}. \tag{S106}$$

Using Lemma S5, we resample according to probabilities $\frac{g^{\phi}_{n,j}}{L^{\phi}_n}$ to see this is properly estimated by

$$L^{\phi}_n \sum_{j=1}^J w^{F,\theta}_{n,j}, \tag{S107}$$

from which we obtain (S105). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left(L^{\phi}_n \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} \right). \tag{S108}$$

Since $w^{F,\theta}_{n,j} = w^{P,\theta}_{n+1,j}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^J w^{P,\theta}_{0,j} = J$ on the denominator and $\sum_{j=1}^J w^{F,\theta}_{N,j}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution, $\nabla_\theta \eta_n^J(h_\theta) \stackrel{a.s.}{\to} \eta_n(\nabla_\theta h_\theta)$, which is not a-priori the gradient of the estimand $\nabla_\theta \eta_n(h_\theta)$. We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all J below.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let $h_\theta : \mathcal{X} \to \mathbb{R}$ be a measurable bounded functional of particles, where $\eta_n^J(h_\theta)$ has two continuous derivatives uniformly bounded over all J by H^* for almost every $\omega \in \Omega$ and $\theta \in \Theta$. If it holds that $\eta_n^J(h_\theta) \stackrel{a.s.}{\to} \eta(h_\theta) = h_\theta^*$, then we also have that $\nabla_\theta \eta_n^J(h_\theta) \stackrel{a.s.}{\to} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*$.*

Proof. Fix $\omega \in \Omega$. The sequence $(\nabla_\theta \eta_n^J(h_\theta)(\omega))_{J\in\mathbb{N}}$ is uniformly bounded over all J, by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of $\eta_n^J(h_\theta)(\omega)|_{\theta=\theta'}$ is also uniformly bounded over all J for almost every $\omega \in \Omega$ and every $\theta' \in \Theta$. A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence $(\eta_n^J(h_\theta)\omega))_{J\in\mathbb{N}}$ is uniformly equicontinuous over θ for almost every $\omega \in \Omega$. Explicitly, for almost every $\omega \in \Omega$ and every $\epsilon > 0$, there exists some $\delta(\omega) > 0$ such that for every $||\theta - \theta'||_\infty < \delta$ and every $J \in \mathbb{N}$ we have that

$$\big|\big|\eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\big|\big|_\infty < \epsilon. \tag{S109}$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence $(\eta_n^J(h_\theta)(\omega))_{J\in\mathbb{N}}$ converges pointwise for $\theta = \phi$ and almost every $\omega \in \Omega$, and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit $\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega)$. Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J\to\infty} \eta_n^J(h_\theta)(\omega). \tag{S110}$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \to \eta_n(h_\theta) = h_\theta^* \tag{S111}$$

for almost every $\omega \in \Omega$. We then have that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \tag{S112}$$

as we wanted. □

Theorem 4 is proved as a corollary of Theorem S2, where we apply $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$, $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$, and then use the continuous mapping theorem.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP-α. The earlier truncation argument is retained below for comparison. The DMOP recursion discounts a time-s derivative contribution by its actual age. We prove the DMOP-0 result below and retain the former direct extension to $\alpha > 0$ for comparison. Section S7 obtains the sharper result by first applying the finite-lag variance calculation and then recovering geometric discounting through an exact mixture identity.

First, we note the following relation between the Dobrushin contraction coefficient and the alpha-mixing coefficients in our context below.

Lemma S9. *Setting X to be the particle collection at time m, and Y at time n, we have that the alpha mixing coefficients,*

$$\int \left| f_{XY}(x, y) - f_X(x)\, f_Y(y) \right| dx\, dy < \alpha, \tag{S113}$$

are bounded by the Dobrushin coefficient, i.e we have

$$\int \left| f_{Y|X}(y|x_1) - f_{Y|X}(y|x_2) \right| dy < \alpha \tag{S114}$$

for all x_1, x_2.

Proof. We rewrite the alpha-mixing assertion as

$$\int \int \left| f_{Y|X}(y|x) - f_Y(y) \right| dy\, f_X(x)\, dx. \tag{S115}$$

We claim that the Dobrushin coefficient implies

$$\int \left| f_{Y|X}(y|x_1) - f_Y(y) \right| dy < \alpha \tag{S116}$$

for all x_1. This is shown as follows:

$$\begin{aligned}
\int \left| f_{Y|X}(y|x) - f_Y(y) \right| dy &= \int \left| \int \left[f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right] f_X(x)\, dx \right| dy && \text{(S117)} \\
&< \int \int \left| f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right| dy\, f_X(x)\, dx && \text{(S118)} \\
&< \int \alpha f_X(x)\, dx && \text{(S119)} \\
&= \alpha && \text{(S120)}
\end{aligned}$$

We then have the desired result. □

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. The time-n derivative contribution is a bounded function of the time-n differentiable particle; this is used to recover the $O(N/J)$ variance bound for DMOP-0. The former truncation extension to $\alpha > 0$ is retained for comparison. We start by decomposing the MOP-α estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} = \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} (w_{n-1,j}^{F,\theta})^{\alpha}}. \tag{S121}$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem S3 (Variance of MOP-0 Gradient Estimate)**.** *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then the variance of DMOP-0, i.e. the algorithm of Naesseth et al. (2018), is*

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^0(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2}{J}. \tag{S122}$$

Proof. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,0} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta,0}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta,0}} \tag{S123}$$

$$= \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^{J} s_{n,j} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^{J} \frac{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{P,\theta}; \theta\big)}{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{P,\phi}; \phi\big)}. \tag{S124}$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{P,\theta}; \theta\big)}{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{P,\phi}; \phi\big)}, \tag{S125}$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^{N} \nabla_\theta \log \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right) \tag{S126}$$

$$= \sum_{n=1}^{N} \frac{\nabla_\theta \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)}{\left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)} \tag{S127}$$

$$= \sum_{n=1}^{N} \frac{\sum_{j=1}^{J} \nabla_\theta s_{n,j}}{\sum_{j=1}^{J} s_{n,j}} \tag{S128}$$

$$= \sum_{n=1}^{N} \frac{1}{J} \sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\theta}; \theta\big)}{f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{F,\phi}; \phi\big)} \tag{S129}$$

$$= \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big), \tag{S130}$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J:

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big). \tag{S131}$$

Therefore, for a given θ_i in the parameter vector θ,

$$\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0(\theta)\right) = \frac{1}{J^2}\mathrm{Var}\left(\sum_{n=1}^{N}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)\Big)\right) \tag{S132}$$

$$= \frac{1}{J^2}\sum_{n=1}^{N}\mathrm{Var}\left(\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\Big)\right)$$

$$+ 2\sum_{m<n}\mathrm{Cov}\left(\frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta};\theta)\Big), \frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial_{\theta_i}}\log\Big(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\Big)\right)$$

$$= \sum_{n=1}^{N}\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right) + 2\sum_{m<n}\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right). \tag{S133}$$

Here, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y\in E}\|M(x,\cdot)-M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu\in\mathcal{P},\mu\neq\nu}\frac{\|\mu M-\nu M\|_{\mathrm{TV}}}{\|\mu-\nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \leq (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor}, \tag{S134}$$

i.e. the mixing time of the particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}, \underline{\mathrm{M}}, \bar{G}, \underline{\mathrm{G}}$ in (A2) and (A3).

Each time-n contribution is a function of the current differentiable particle cloud. Lemma 2.3 of Karjalainen et al. (2023), followed by conditional multinomial resampling and the bounded Bayes-ratio step, gives the centered L^4 bound below. Theorem 3.1 of the same paper gives the displayed particle-cloud mixing coefficient. At $\alpha = 0$ these two results give

$$\left\|\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta) - \mathbb{E}\left[\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right]\right\|_{L^4} \leq \frac{CG'(\theta)}{\sqrt{J}},$$
$$a(k) \leq (1-\epsilon)^{\lfloor (k-1)/[c\{1+\log J\}]\rfloor}. \tag{S135}$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S135) gives

$$\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right) \leq (1-\epsilon)^{\frac{1}{2}\left\lfloor\frac{|n-m|-1}{c\{1+\log(J)\}}\right\rfloor}\frac{G'(\theta)^2}{J}. \tag{S136}$$

Putting it all together, we see that

$$\operatorname{Var}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}^0(\theta)\right) = \sum_{n=1}^{N}\operatorname{Var}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right) + 2\sum_{m<n}\operatorname{Cov}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right) \tag{S137}$$

$$\leq \frac{NG'(\theta)^2}{J} + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\frac{G'(\theta)^2}{J} \tag{S138}$$

$$\leq \frac{G'(\theta)^2}{J}\left(N + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\right) \tag{S139}$$

$$\lesssim \frac{G'(\theta)^2 N}{J}. \tag{S140}$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\operatorname{Var}\big(\nabla_\theta\hat{\ell}^0(\theta)\big) \lesssim \frac{Np\,G'(\theta)^2}{J}. \tag{S141}$$

□

Note that the factor of N that pops up here is due to the use of the unnormalized gradient. If one divides the gradient estimate by $\sqrt{N}$ before usage, the variance does not depend on the horizon. If one divides by N, the error is $O(1/\sqrt{NJ})$.

The preceding calculation gives the order $NpG'(\theta)^2/J$ for the DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of α.

Theorem S4 (MSE and variance bounds for DMOP-α)**.** *Suppose the observations are fixed, $J \geq 2$, $\alpha \in [0,1]$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then*

$$\max\left\{\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell^\alpha(\theta)\right\|_2^2, \operatorname{Var}\left\{\nabla_\theta\hat{\ell}^\alpha(\theta)\right\}\right\} \lesssim pG'(\theta)^2\left(\frac{N}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{S142}$$

For $0 \leq \alpha < 1$,

$$\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2\left[\left(\frac{N}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2 + \left\{\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q\right\}^2\right]$$

$$\lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} + N^2\left\{\frac{\varrho(1-\alpha)}{1-\alpha\varrho}\right\}^2\right]. \tag{S143}$$

For the same range of α, the population component of the fixed-observation bias satisfies

$$\|\nabla_\theta\ell^\alpha(\theta) - \nabla_\theta\ell(\theta)\|_2 \leq CN\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}. \tag{S144}$$

Here $\varrho \in (0,1)$. Apart from the particle-system mixing and logarithmic-threshold factor hidden by $\lesssim$, the suppressed constants depend only on the fixed model bounds and not on N, J, or α. Section S7 gives the complete proof. The particle argument first bounds each finite lag truncation and then uses the exact geometric-mixture identity; it does not replace DMOP-α by a hard-cutoff estimator.

Proof. Lemma S13 proves Equation (S142). Theorem S5 proves Equation (S143), and Lemma S14 proves Equation (S144). □

S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S4. Write $\nabla_\theta \ell_n^\alpha(\theta)$ for the nth conditional score produced by the deterministic differentiated population recursion. At $\theta = \phi$, its $\alpha = 1$ version is the exact conditional score $\nabla_\theta \ell_n(\theta)$. Along each selected lineage, the log-weight gradient obeys the existing recursion

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \alpha\, \nabla_\theta \log w_{n-1,k_j}^{F,\theta}\Big|_{\theta=\phi} + \nabla_\theta^{\mathrm{tot}} \log g_{n,k_j}^{\theta}\Big|_{\theta=\phi}. \tag{S145}$$

The final term is the one-time total measurement gradient, denoted by h_n and defined explicitly in Equation (S152) below. Equation (S145) is the derivative at $\theta = \phi$ of the log-space recursion

$$\begin{aligned}
\log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta}, \\
\log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^{\theta} - \log g_{n,k_j}^{\phi}.
\end{aligned} \tag{S146}$$

At $\theta = \phi$, the log weights are zero but their gradients need not be. Iterating the recursion gives the discounted sum in Equation (S151). The proof first applies the finite-window particle estimates to each analytical lag truncation and then recovers this discounted sum through an exact geometric-mixture identity. Throughout this section, the observations are fixed. Expectations and variances are over the simulator and particle-filter random variables, conditional on those observations.

We use Assumptions (A3) and (A5) in the precise pathwise sense already required by the differentiated MOP construction. For fixed simulator random numbers and fixed resampling indices, Assumption (A5) says that the recursively simulated state path and the composed log measurement densities are continuously differentiable in θ near ϕ, and that the joint law of the base simulator random numbers does not depend on θ. The derivative in the last clause of Assumption (A3) is the resulting total derivative: uniformly in n, j and in a neighborhood of ϕ, after enlarging $G'(\theta)$ to bound that neighborhood if necessary,

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \right\|_\infty \le G'(\theta). \tag{S147}$$

These two existing assumptions also justify differentiation under the finite-horizon simulator expectations used below. Indeed, the product rule and Equation (S147) give, pathwise,

$$\begin{aligned}
&\left\| \nabla_\theta \prod_{s=1}^{t} G_s(X_s) \right\|_\infty \\
&\quad = \left\| \prod_{s=1}^{t} G_s(X_s) \sum_{s=1}^{t} \nabla_\theta \log G_s(X_s) \right\|_\infty \\
&\le \prod_{s=1}^{t} G_s(X_s) \sum_{s=1}^{t} \left\| \nabla_\theta \log G_s(X_s) \right\|_\infty \\
&\le t \bar{G}^t G'(\theta).
\end{aligned} \tag{S148}$$

The final expression is deterministic and finite for each fixed t. For each coordinate of θ, the mean-value theorem therefore bounds every difference quotient about ϕ by the same quantity $t\bar{G}^t G'(\theta)$. The dominated convergence theorem gives

$$\nabla_\theta \mathbb{E}_U \left[\prod_{s=1}^{t} G_s(X_s) \right] = \mathbb{E}_U \left[\nabla_\theta \prod_{s=1}^{t} G_s(X_s) \right]. \tag{S149}$$

Thus the required interchange follows from the clarified Assumptions (A3) and (A5); it is not an additional assumption. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2). Lemma S11 gives a uniform weak-error bound for a current filtering functional. Its proof is not used for a historical functional carried by the genealogy; that separate issue is audited below.

For the functionals below, evaluated at $\theta = \phi$, let ρ_t be the likelihood-tilted law of the complete fixed-random-number simulator history through time t, including the base simulator variables. Its $X_{0:t}$-marginal is the usual posterior path law. Thus, for every integrable functional h of that complete history,

$$\rho_t(h) = \frac{\mathbb{E}\left[h \prod_{s=1}^{t} G_s(X_s)\right]}{\mathbb{E}\left[\prod_{s=1}^{t} G_s(X_s)\right]}. \tag{S150}$$

Here $G_s(X_s)$ is the existing measurement potential. Future base simulator variables are fresh and conditionally independent of the complete past given the current particle, as in the original POMP construction.

Unrolling the existing filtered-weight recursion at $\theta = \phi$ gives

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \; \nabla_\theta \log g_{n-r,j}^{A,F,\theta}\Big|_{\theta=\phi}. \tag{S151}$$

For a complete ancestral path, write h_k, only in this section, for the measurement-score contribution

$$h_k = \nabla_\theta \log g_k^{A,F,\theta}\Big|_{\theta=\phi} \tag{S152}$$

carried on that path. In the path-space notation of Chan et al. (2018), the notation used below means explicitly

$$m(\xi_t) = \frac{1}{J}\sum_{j=1}^{J} \delta_{\xi_{t,j}}, \qquad m(\xi_t)(h_k) = \frac{1}{J}\sum_{j=1}^{J} h_k(\xi_{t,j}), \quad 1 \le k \le t, \tag{S153}$$

where $\xi_{t,j}$ is the complete ancestral particle path carried by particle j at time t. Thus $h_k(\xi_{t,j})$ is the single time-k measurement-score contribution on that particle, not an arbitrary functional denoted by m. Equation (S150) gives its population counterpart.

We now identify the lag coefficients in the conditional score. At $\theta = \phi$, all weights equal one, so differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\nabla_\theta \hat{\ell}_n^{\alpha}(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} - \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta}\Big|_{\theta=\phi}$$

$$= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} - \frac{\alpha}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi}, \tag{S154}$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha$. In the MOP-$\alpha$ recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time-n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta}\frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \tag{S155}$$

Consequently, the second average in Equation (S154) is pathwise

$$\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta}\Big|_{\theta=\phi} = \frac{\alpha}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi}, \tag{S156}$$

which is the empirical average over the time-$(n-1)$ filtered genealogy. Substituting Equation (S151) into Equation (S154) and changing the index in the second sum give

$$\begin{aligned}\nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \left\{ m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}) \right\}. \end{aligned}\tag{S157}$$

Consequently, define

$$\begin{aligned} \Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \le r < n. \end{aligned}\tag{S158}$$

We now verify that the right-hand population quantities are the lag coefficients of the deterministic differentiated recursion. Equations (S149) and (S152) give, at $\theta = \phi$,

$$\begin{aligned} &\nabla_\theta \log \mathbb{E}_U\left[\prod_{s=1}^{n} G_s(X_s)\right] \\ &\quad = \frac{\mathbb{E}_U\left[\prod_{s=1}^n G_s(X_s) \sum_{s=1}^n h_s\right]}{\mathbb{E}_U\left[\prod_{s=1}^n G_s(X_s)\right]} \\ &\quad = \sum_{s=1}^{n} \rho_n(h_s). \end{aligned}\tag{S159}$$

Subtracting the same identity at time $n-1$ gives

$$\begin{aligned} \nabla_\theta \ell_n(\theta) &= \sum_{s=1}^{n} \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \left\{ \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \right\} \end{aligned}$$

$$= \sum_{r=0}^{n-1} \Delta_{n,r}. \tag{S160}$$

Finally, differentiating the two population weighted averages in the deterministic MOP-α recursion and using Equation (S149) gives

$$\begin{aligned}
\nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \left\{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\right\} \\
&= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}.
\end{aligned} \tag{S161}$$

Lemma S10 (Lagwise first-derivative forgetting)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0,1)$ such that, for $0 \le r < n$,*

$$\|\Delta_{n,r}\|_2 \le C\sqrt{p}\, G'(\theta)\varrho^r. \tag{S162}$$

Proof. We derive the result from the posterior update and the contraction of the posterior backward kernels.

Step 1: one-step posterior update. First suppose that $1 \le r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \ldots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of h_k in Equation (S152). By Equation (S147),

$$|h_{k,i}| \le G'(\theta). \tag{S163}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \le n-1$, Equation (S150) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}
\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U\left[h_{k,i} \prod_{s=1}^{n-1} G_s(X_s)\mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\}\right]}{\mathbb{E}_U\left[\prod_{s=1}^{n-1} G_s(X_s)\mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\}\right]} \\
&= \frac{\rho_{n-1}\{h_{k,i} M_n(G_n)\}}{\rho_{n-1} M_n(G_n)},
\end{aligned} \tag{S164}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n-1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U\{\prod_{s=1}^{n-1} G_s(X_s)\}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n). \tag{S165}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \ge \underline{\mathrm{G}} \int M_n(x_{n-1}, dx_n) = \underline{\mathrm{G}}, \tag{S166}$$

$$M_n(G_n)(x_{n-1}) \le \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S167}$$

Step 2: backward-kernel contraction. We next control how much information about the path through time k can remain at time $n-1$. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1}, denoted below by B_s, is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S168}$$

Assumption (A2) gives the following minorization for every measurable set A:

$$\begin{aligned} B_s(x_s, A) &= \frac{\displaystyle\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)} \\ &\geq \frac{\underline{\mathrm{M}}\, \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \tag{S169}$$

Put $\epsilon = \underline{\mathrm{M}}/\bar{M}$. If $\epsilon < 1$, Equation (S169) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1-\epsilon) R_s(x_s, \cdot), \tag{S170}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1-\epsilon} \tag{S171}$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x_s',

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x_s') &= (1-\epsilon)\{R_s(f)(x_s) - R_s(f)(x_s')\}, \\ \left|B_s(f)(x_s) - B_s(f)(x_s')\right| &\leq (1-\epsilon)\operatorname{osc}(f). \end{aligned} \tag{S172}$$

If $\epsilon = 1$, Equation (S169) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x_s' gives

$$\operatorname{osc}\{B_s(f)\} \leq (1-\epsilon)\operatorname{osc}(f). \tag{S173}$$

Step 3: project and propagate the time-k contribution. The total derivative $h_{k,i}$ may depend on the complete simulator genealogy through time k. Its conditional projection onto the physical location is

$$\bar{h}_{k,i}(X_k) = \mathbb{E}_{\rho_k}(h_{k,i} \mid X_k). \tag{S174}$$

We first verify that the same projection applies under every later posterior path law. For $t \geq k$, let

$$L_{k,t}(X_k) = \mathbb{E}_U \left\{ \prod_{s=k+1}^{t} G_s(X_s) \,\middle|\, X_k \right\}, \qquad L_{k,k} = 1. \tag{S175}$$

For every bounded function q of X_k, the Markov property and Equation (S150) give

$$\rho_t\{h_{k,i} q(X_k)\} = \frac{\mathbb{E}_U\left[h_{k,i} q(X_k) \prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k)\right]}{\mathbb{E}_U\left[\prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k)\right]}$$

$$
\begin{aligned}
&= \frac{\mathbb{E}_U\left[\bar{h}_{k,i}(X_k)q(X_k)\prod_{s=1}^k G_s(X_s)L_{k,t}(X_k)\right]}{\mathbb{E}_U\left[\prod_{s=1}^k G_s(X_s)L_{k,t}(X_k)\right]} \\
&= \rho_t\{\bar{h}_{k,i}(X_k)q(X_k)\}.
\end{aligned} \tag{S176}
$$

The second equality is the defining conditional-expectation identity in Equation (S174), multiplied by the X_k-measurable function $q(X_k)L_{k,t}(X_k)$. Hence

$$
\mathbb{E}_{\rho_t}(h_{k,i} \mid X_k) = \bar{h}_{k,i}(X_k), \qquad t \geq k. \tag{S177}
$$

Repeated conditioning through the physical backward kernels therefore gives

$$
\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = x_{n-1}\} = (B_{n-1}B_{n-2}\cdots B_{k+1}\bar{h}_{k,i})(x_{n-1}). \tag{S178}
$$

There are $n-1-k = r-1$ backward kernels in this display. Moreover, Equation (S163) gives

$$
\begin{aligned}
\|\bar{h}_{k,i}\|_\infty &\leq \|h_{k,i}\|_\infty \leq G'(\theta), \\
\operatorname{osc}(\bar{h}_{k,i}) &\leq 2\|\bar{h}_{k,i}\|_\infty \\
&\leq 2G'(\theta).
\end{aligned} \tag{S179}
$$

If $r = 1$, Equation (S178) contains no backward kernel. If $r \geq 2$, applying Equation (S173) successively gives

$$
\begin{aligned}
&\operatorname{osc}\left[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}\right] \\
&\quad \leq 2G'(\theta)\begin{cases} 1, & r = 1, \\ (1-\epsilon)^{r-1}, & r \geq 2. \end{cases}
\end{aligned} \tag{S180}
$$

Step 4: covariance identity and geometric bound. To finish, abbreviate the conditional expectation on the left of Equation (S180) by H, only for the next calculation. Since the X_{n-1}-marginal of ρ_{n-1} is π_{n-1}, the tower property gives

$$
\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \tag{S181}
$$

$$
\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \tag{S182}
$$

Substituting Equations (S181)– (S182) into Equation (S164) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$
\begin{aligned}
\Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\
&= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)} \\
&= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}.
\end{aligned} \tag{S183}
$$

For any probability measure μ and bounded real f,

$$
|f(x) - \mu(f)| \leq \operatorname{osc}(f). \tag{S184}
$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix $x^\star$. Assumption (A2) gives, for every measurable set A,

$$\begin{aligned} M_n(x, A) &= \int_A f_{X_n|X_{n-1}}(x_n \mid x; \theta)\, dx_n \\ &\geq \underline{\mathrm{M}} \int_A dx_n \\ &\geq \frac{\underline{\mathrm{M}}}{\bar{M}} \int_A f_{X_n|X_{n-1}}(x_n \mid x^\star; \theta)\, dx_n \\ &= \epsilon M_n(x^\star, A). \end{aligned} \tag{S185}$$

If $\epsilon < 1$, Equation (S185) permits the decomposition

$$M_n(x, \cdot) = \epsilon M_n(x^\star, \cdot) + (1-\epsilon) R_n(x, \cdot), \tag{S186}$$

where R_n is a Markov kernel. Therefore, for any x, x',

$$\begin{aligned} M_n(G_n)(x) - M_n(G_n)(x') &= (1-\epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\}, \\ \left|M_n(G_n)(x) - M_n(G_n)(x')\right| &\leq (1-\epsilon) \operatorname{osc}(G_n) \\ &\leq (1-\epsilon)(\bar{G} - \underline{\mathrm{G}}). \end{aligned} \tag{S187}$$

Taking the supremum over x, x' gives

$$\operatorname{osc}\{M_n(G_n)\} \leq (1-\epsilon)(\bar{G} - \underline{\mathrm{G}}). \tag{S188}$$

If $\epsilon = 1$, Equation (S185) implies that $M_n(x, \cdot)$ does not depend on x, so the left side of Equation (S188) is zero and the same bound holds. Taking absolute values in Equation (S183), using Equations (S184)– (S188), and then using Equation (S180), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\operatorname{osc}(H) \operatorname{osc}\{M_n(G_n)\}}{\underline{\mathrm{G}}} \\ &\leq \frac{2(\bar{G} - \underline{\mathrm{G}})}{\underline{\mathrm{G}}} G'(\theta) \begin{cases} (1-\epsilon), & r = 1, \\ (1-\epsilon)^r, & r \geq 2. \end{cases} \end{aligned} \tag{S189}$$

For both cases, take

$$\varrho = 1 - \frac{\epsilon}{2} = 1 - \frac{\underline{\mathrm{M}}}{2\bar{M}} \in (0, 1). \tag{S190}$$

Since $1 - \epsilon \leq \varrho$, the two cases in Equation (S189) are both bounded by ϱ^r. Summing the squared coordinate bounds and taking square roots give

$$\begin{aligned} \|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^{p} |\Delta_{n,r,i}|^2 \right\}^{1/2} \\ &\leq C\sqrt{p}\, G'(\theta) \varrho^r, \qquad 1 \leq r < n. \end{aligned} \tag{S191}$$

Step 5: the zero-lag case. Finally, when $r = 0$, Equation (S158) and Equation (S147) give

$$\|\Delta_{n,0}\|_2 = \|\rho_n(h_n)\|_2$$

$$
\begin{aligned}
&\leq \rho_n(\|h_n\|_2) \\
&\leq \sqrt{p}\, G'(\theta) \\
&\leq C\sqrt{p}\, G'(\theta)\varrho^0. \qquad \text{(S192)}
\end{aligned}
$$

Equations (S191) and (S192) prove Equation (S162). If $\epsilon = 1$, Equation (S183) and the zero oscillation in Equation (S188) show that $\Delta_{n,r} = 0$ for every $r \geq 1$, so any fixed $\varrho \in (0,1)$ proves the same result. Thus ϱ depends only on the model minorization bounds and is independent of N, J, α, and any lag truncation k. In particular, the factor ϱ^r includes the contraction of the first positive lag; it is not only a tail bound. The projection in Equation (S174) is sufficient here because $\Delta_{n,r}$ is a difference of two linear expectations of one historical input. It does not eliminate conditional covariance between two genealogical inputs, and therefore does not supply the particle covariance bound that would be needed for the sharper particle rate.

□

Lemma S11 (Uniform weak bias of current-particle averages)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is $C < \infty$ such that, for every bounded real function f of the current particle location and every $t \geq 1$,*

$$
|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq \frac{C}{J}\operatorname{osc}(f). \qquad \text{(S193)}
$$

The constant is independent of t and J. Here "current particle" includes the particle value differentiated with respect to θ. In particular, the bounded time-t contribution h_t in Equation (S152) is an admissible choice of f at the time at which it is created.

Proof. The proof has four stages: reduce the filtered average to the prediction cloud, establish contraction of the normalized semigroup, compute the J^{-1} bias of one particle update, and telescope those updates.

Step 1: prediction-to-filtering reduction. The predictor cloud consists of the current particles. A bounded function of the current differentiable particle is therefore a current-state functional in the Feynman–Kac model already defined in Section S3. The current-location marginal of ρ_t is the filtering distribution, and hence

$$
\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \qquad \text{(S194)}
$$

Conditionally on the prediction cloud η_t^J, multinomial resampling draws the filtering particles independently from $\Psi_t(\eta_t^J)$. Therefore

$$
\begin{aligned}
\mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} &= \frac{1}{J}\sum_{j=1}^{J} \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\} \\
&= \frac{1}{J}\sum_{j=1}^{J} \Psi_t(\eta_t^J)(f) \\
&= \Psi_t(\eta_t^J)(f). \qquad \text{(S195)}
\end{aligned}
$$

Step 2: normalized-semigroup contraction. Take $\Phi_{t,t}$ to be the identity. For $1 \leq p \leq t$, define

$$
Q_{t,t}(f)(x) = G_t(x)f(x),
$$

$$Q_{p,t}(f)(x) = G_p(x) M_{p+1}\{Q_{p+1,t}(f)\}(x), \qquad p < t. \tag{S196}$$

The definitions of $\Phi_{p,t}$ and Ψ_t give, for every probability measure μ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \tag{S197}$$

For $p < t$,

$$Q_{p,t}(1)(x) = G_p(x) \int f_{X_{p+1}|X_p}(u \mid x; \theta) Q_{p+1,t}(1)(u)\, du. \tag{S198}$$

Thus, for every x, x',

$$\begin{aligned} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\le \frac{\bar{G}\bar{M} \int Q_{p+1,t}(1)(u)\, du}{\underline{\mathrm{GM}} \int Q_{p+1,t}(1)(u)\, du} \\ &= \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}}. \end{aligned} \tag{S199}$$

For $p = t$, the same ratio is at most $\bar{G}/\underline{\mathrm{G}}$. Put

$$R = \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}} \ge 1. \tag{S200}$$

Then

$$\sup_{x,x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \le R, \qquad 1 \le p \le t. \tag{S201}$$

For $p \le s < t$, the normalized forward kernel is

$$P^{(t)}_{s,s+1}(x, A) = \frac{\displaystyle\int_A f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}{\displaystyle\int f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}. \tag{S202}$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\displaystyle\int_A Q_{s+1,t}(1)(u)\, du}{\displaystyle\int Q_{s+1,t}(1)(u)\, du}. \tag{S203}$$

Assumption (A2) gives

$$\begin{aligned} P^{(t)}_{s,s+1}(x, A) &\ge \frac{\underline{\mathrm{M}} \displaystyle\int_A Q_{s+1,t}(1)(u)\, du}{\bar{M} \displaystyle\int Q_{s+1,t}(1)(u)\, du} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \nu_{s,t}(A). \end{aligned} \tag{S204}$$

Consequently, by Equation (S190), the Dobrushin coefficient of every kernel in Equation (S202) is at most ϱ. The factor $G_s(x)$ cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P^{(t)}_{p,p+1}P^{(t)}_{p+1,p+2}\cdots P^{(t)}_{t-1,t}f. \tag{S205}$$

It follows that

$$\operatorname{osc}\left\{\frac{Q_{p,t}(f)}{Q_{p,t}(1)}\right\} \le \varrho^{t-p}\operatorname{osc}(f). \tag{S206}$$

Step 3: one complete particle update. We next bound one complete selection–mutation update. For $p = 1$, put $\mu = \eta_1 = \pi_0 M_1$. For $p \ge 2$, condition on η^J_{p-1} and put

$$\mu = \Phi_p(\eta^J_{p-1}). \tag{S207}$$

In either case, the particles forming η^J_p are conditionally independent with common law μ. Here $Q_{p,t}$ propagates the local error to time t: q isolates its random normalizer, while d is the centered empirical fluctuation. Within this calculation, put

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \qquad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \qquad d = \frac{Q_{p,t}(f-a)}{\mu Q_{p,t}(1)}. \tag{S208}$$

Then

$$\mu(q) = 1, \qquad \mu(d) = 0, \tag{S209}$$

and Equation (S197) gives

$$\begin{aligned}
&\Psi_t\{\Phi_{p,t}(\eta^J_p)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f)\\
&= \frac{\eta^J_p Q_{p,t}(f)}{\eta^J_p Q_{p,t}(1)} - a\\
&= \frac{\eta^J_p Q_{p,t}(f-a)}{\eta^J_p Q_{p,t}(1)}\\
&= \frac{\eta^J_p(d)}{\eta^J_p(q)}.
\end{aligned} \tag{S210}$$

Equation (S201) implies

$$R^{-1} \le q(x) \le R, \qquad \eta^J_p(q) \ge R^{-1}, \qquad \|q-1\|_\infty \le R. \tag{S211}$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \tag{S212}$$

we have $a = \mu(qF)$ and $\mu(q) = 1$. Thus a is an average of F under the probability measure $q\,d\mu$, so

$$\inf F \le a \le \sup F. \tag{S213}$$

Since $d = q(F-a)$, Equations (S211), (S213), and (S206) give

$$\|d\|_\infty \le R\operatorname{osc}(F)$$

$$\leq R\varrho^{t-p}\,\mathrm{osc}(f). \tag{S214}$$

Conditional independence and $\mu(d)=0$ give

$$\begin{aligned}\mathbb{E}\{\eta_p^J(d)\mid\mu\}&=0,\\ \mathbb{E}\{\eta_p^J(d)^2\mid\mu\}&=\frac{1}{J}\mu(d^2),\\ \mathbb{E}\{(\eta_p^J(q)-1)^2\mid\mu\}&=\frac{1}{J}\mu\{(q-1)^2\}.\end{aligned} \tag{S215}$$

Consequently,

$$\begin{aligned}&\mathbb{E}\left\{\left.\frac{\eta_p^J(d)}{\eta_p^J(q)}\right|\mu\right\}\\ &\quad=-\mathbb{E}\left\{\left.\frac{\eta_p^J(d)\{\eta_p^J(q)-1\}}{\eta_p^J(q)}\right|\mu\right\}.\end{aligned} \tag{S216}$$

Conditional Cauchy–Schwarz and Equations (S211)– (S215) yield

$$\begin{aligned}&\left|\mathbb{E}\left\{\left.\frac{\eta_p^J(d)}{\eta_p^J(q)}\right|\mu\right\}\right|\\ &\leq R\left[\mathbb{E}\{\eta_p^J(d)^2\mid\mu\}\mathbb{E}\{(\eta_p^J(q)-1)^2\mid\mu\}\right]^{1/2}\\ &=\frac{R}{J}\left[\mu(d^2)\mu\{(q-1)^2\}\right]^{1/2}\\ &\leq\frac{C}{J}\varrho^{t-p}\,\mathrm{osc}(f).\end{aligned} \tag{S217}$$

Step 4: telescope over update times. It remains to write the exact full-update telescope. Since $\eta_t=\Phi_{1,t}(\eta_1)$,

$$\begin{aligned}&\mathbb{E}\Psi_t(\eta_t^J)(f)-\Psi_t(\eta_t)(f)\\ &\quad=\mathbb{E}\left[\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f)-\Psi_t\{\Phi_{1,t}(\eta_1)\}(f)\right]\\ &\qquad+\sum_{p=2}^{t}\mathbb{E}\left[\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f)-\Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)\right].\end{aligned} \tag{S218}$$

Indeed, for $2\leq p\leq t$,

$$\Phi_{p,t}\circ\Phi_p=\Phi_{p-1,t}, \tag{S219}$$

so consecutive terms cancel. The only terms left are $\Psi_t(\eta_t^J)(f)$ and $-\Psi_t(\eta_t)(f)$.

Apply Equation (S217) to every summand in Equation (S218). Then

$$\begin{aligned}\left|\mathbb{E}\Psi_t(\eta_t^J)(f)-\Psi_t(\eta_t)(f)\right|&\leq\frac{C\,\mathrm{osc}(f)}{J}\sum_{p=1}^{t}\varrho^{t-p}\\ &\leq\frac{C}{J(1-\varrho)}\,\mathrm{osc}(f).\end{aligned} \tag{S220}$$

Absorb $(1-\varrho)^{-1}$ into C and combine this display with Equations (S194) and (S195). $\square$

Lemma S12 (Full-score summation-by-parts identities)**.** *Suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then the particle and population scores satisfy Equations (S224) and (S225) below. Moreover, for every coordinate i and $t \leq N$,*

$$\left\| \sum_{k=1}^{t} \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^{t} \alpha^{t-k}. \tag{S221}$$

Proof. Step 1: particle summation by parts. We first record the summation-by-parts identity used again in the variance proof. Sum Equation (S157) over $n = 1, \ldots, N$ and put $k = n - r$. Since the sums are finite,

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = \sum_{k=1}^{N} \left[m(\xi_k)(h_k) + \sum_{t=k+1}^{N} \alpha^{t-k} \{ m(\xi_t)(h_k) - m(\xi_{t-1})(h_k) \} \right]. \tag{S222}$$

For each fixed k, collecting the coefficient of every $m(\xi_t)(h_k)$ gives

$$\begin{aligned} m(\xi_k)(h_k) &+ \sum_{t=k+1}^{N} \alpha^{t-k} \{ m(\xi_t)(h_k) - m(\xi_{t-1})(h_k) \} \\ &= (1 - \alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \end{aligned} \tag{S223}$$

Indeed, the coefficients are $1 - \alpha$ at $t = k$, $\alpha^{t-k} - \alpha^{t+1-k} = (1 - \alpha)\alpha^{t-k}$ for $k < t < N$, and α^{N-k} at $t = N$. Interchanging the finite sums once more yields

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} m(\xi_t) \left(\sum_{k=1}^{t} \alpha^{t-k} h_k \right) + m(\xi_N) \left(\sum_{k=1}^{N} \alpha^{N-k} h_k \right). \tag{S224}$$

Step 2: population identity and uniform bound. The same algebra applied to Equation (S161) gives

$$\nabla_\theta \ell^\alpha(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} \rho_t \left(\sum_{k=1}^{t} \alpha^{t-k} h_k \right) + \rho_N \left(\sum_{k=1}^{N} \alpha^{N-k} h_k \right). \tag{S225}$$

For every coordinate i, Equation (S147) gives

$$\left\| \sum_{k=1}^{t} \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^{t} \alpha^{t-k}. \tag{S226}$$

Equation (S226) is identical to Equation (S221), completing the proof. □

Claim S1 (Conditional empirical moments)**.** *Let $\mathcal{F}$ be a sigma-field and let μ be an $\mathcal{F}$-measurable probability measure. Suppose that, conditionally on $\mathcal{F}$, $X_1, \ldots, X_J$ are independent with common law μ. The following bounds hold for bounded $\mathcal{F}$-measurable real functions f, f_1, f_2:*

$$\mathbb{E} \left\{ \frac{1}{J} \sum_{j=1}^{J} \{ f(X_j) - \mu(f) \} \middle| \mathcal{F} \right\} = 0, \tag{S227}$$

$$\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^2\middle|\mathcal{F}\right\}=\frac{\mu\{(f-\mu f)^2\}}{J}\le\frac{\operatorname{osc}(f)^2}{4J}, \tag{S228}$$

$$\begin{aligned}\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^4\middle|\mathcal{F}\right\}&=\frac{\mu\{(f-\mu f)^4\}}{J^3}+\frac{3(J-1)}{J^3}\mu\{(f-\mu f)^2\}^2\\&\le\frac{\mu\{(f-\mu f)^4\}}{J^3}+\frac{3}{J^2}\mu\{(f-\mu f)^2\}^2\\&\le\frac{3\operatorname{osc}(f)^4}{J^2}.\end{aligned} \tag{S229}$$

Consequently, for bounded f_1 and f_2,

$$\begin{aligned}&\left\|\left[\frac{1}{J}\sum_{j=1}^{J}\{f_1(X_j)-\mu(f_1)\}\right]\left[\frac{1}{J}\sum_{j=1}^{J}\{f_2(X_j)-\mu(f_2)\}\right]\right\|_{L^2|\mathcal{F}}\\&\quad\le\frac{\sqrt{3}}{J}\operatorname{osc}(f_1)\operatorname{osc}(f_2).\end{aligned} \tag{S230}$$

Proof. Put $Y_j=f(X_j)-\mu(f)$. Conditional independence and $\mathbb{E}(Y_j\mid\mathcal{F})=0$ give Equations (S227) and (S228). They also give

$$\begin{aligned}&\mathbb{E}\left\{\left(\frac{1}{J}\sum_{j=1}^{J}Y_j\right)^4\middle|\mathcal{F}\right\}\\&=\frac{1}{J^4}\left\{J\mu\{(f-\mu f)^4\}+3J(J-1)\mu\{(f-\mu f)^2\}^2\right\}\\&=\frac{\mu\{(f-\mu f)^4\}}{J^3}+\frac{3(J-1)}{J^3}\mu\{(f-\mu f)^2\}^2\\&\le\frac{3}{J^2}\operatorname{osc}(f)^4,\end{aligned} \tag{S231}$$

which is Equation (S229). Conditional Cauchy–Schwarz applied to the two fourth moments proves Equation (S230). □

Claim S2 (Normalized-ratio perturbations)**.** *Let μ and ν be probability measures. For $l\in\{0,1\}$, let $a_l>0$ and b_l be bounded functions such that $\mu(a_l)=1$, and put*

$$\mathcal{R}_l(\lambda)=\frac{\lambda(b_l)}{\lambda(a_l)},\qquad D_\mu\mathcal{R}_l=b_l-\mathcal{R}_l(\mu)a_l. \tag{S232}$$

Then

$$\mathcal{R}_l(\nu)-\mathcal{R}_l(\mu)=(\nu-\mu)D_\mu\mathcal{R}_l-\frac{(\nu-\mu)(a_l)}{\nu(a_l)}(\nu-\mu)D_\mu\mathcal{R}_l, \tag{S233}$$

and hence

$$\{\mathcal{R}_1(\nu)-\mathcal{R}_0(\nu)\}-\{\mathcal{R}_1(\mu)-\mathcal{R}_0(\mu)\}$$

$$
\begin{aligned}
&- (\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\} \\
&= -\frac{(\nu-\mu)(a_1-a_0)}{\nu(a_0)\nu(a_1)}(\nu-\mu)D_\mu\mathcal{R}_1 \\
&\quad - \frac{(\nu-\mu)(a_0)}{\nu(a_0)}(\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\}.
\end{aligned} \tag{S234}
$$

If $\inf_{l\in\{0,1\}}\nu(a_l) \geq C^{-1}$*, then the absolute value of the left side of Equation (S234) is at most*

$$
\begin{aligned}
&C^2\left|(\nu-\mu)(a_1-a_0)\right|\left|(\nu-\mu)D_\mu\mathcal{R}_1\right| \\
&\quad + C\left|(\nu-\mu)(a_0)\right|\left|(\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\}\right|.
\end{aligned} \tag{S235}
$$

Proof. Since $\mu(a_l) = 1$,

$$
\begin{aligned}
\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) &= \frac{\nu(b_l) - \mathcal{R}_l(\mu)\nu(a_l)}{\nu(a_l)} \\
&= \frac{(\nu-\mu)D_\mu\mathcal{R}_l}{\nu(a_l)} \\
&= (\nu-\mu)D_\mu\mathcal{R}_l - \frac{(\nu-\mu)(a_l)}{\nu(a_l)}(\nu-\mu)D_\mu\mathcal{R}_l,
\end{aligned} \tag{S236}
$$

which proves Equation (S233). Moreover,

$$
\begin{aligned}
&\frac{(\nu-\mu)(a_1)}{\nu(a_1)} - \frac{(\nu-\mu)(a_0)}{\nu(a_0)} \\
&= \left(1 - \frac{1}{\nu(a_1)}\right) - \left(1 - \frac{1}{\nu(a_0)}\right) \\
&= \frac{\nu(a_1) - \nu(a_0)}{\nu(a_0)\nu(a_1)} \\
&= \frac{(\nu-\mu)(a_1-a_0)}{\nu(a_0)\nu(a_1)}.
\end{aligned} \tag{S237}
$$

Subtracting the cases $l = 0$ and $l = 1$ in Equation (S233) now proves Equation (S234). Taking absolute values there and using $\nu(a_0)^{-1} \leq C$ and $\nu(a_1)^{-1} \leq C$ proves Equation (S235). This exact identity is the paired specialization of the normalized Feynman–Kac perturbation calculation in Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011). □

Lemma S13 (Particle error and variance of the DMOP-α score)**.** *Suppose* $J \geq 2$, $\theta = \phi$, $\alpha \in [0,1]$, *Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then*

$$
\operatorname{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} \lesssim \frac{NpG'(\theta)^2}{J}\left(\sum_{r=0}^{N-1}\alpha^r\right)^2, \tag{S238}
$$

$$
\left\|\mathbb{E}\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell^\alpha(\theta)\right\|_2 \lesssim \frac{N\sqrt{p}\,G'(\theta)}{J}\sum_{r=0}^{N-1}\alpha^r, \tag{S239}
$$

$$
\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell^\alpha(\theta)\right\|_2^2 \lesssim pG'(\theta)^2\left(\frac{N}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{S240}
$$

The finite geometric sum is

$$\sum_{r=0}^{N-1} \alpha^r = \begin{cases} \dfrac{1-\alpha^N}{1-\alpha}, & 0 \le \alpha < 1, \\ N, & \alpha = 1. \end{cases} \tag{S241}$$

As in the earlier variance results, $\lesssim$ hides the particle-system mixing and logarithmic-threshold factor. Apart from that factor, the suppressed constants depend only on the fixed model bounds and are independent of $N, J,$ and α. As elsewhere, $\operatorname{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$ *for a random vector Z.*

Proof. No hard cutoff is used by the algorithm. The proof uses lag truncations only as an analytical decomposition of the geometrically discounted score. It first proves a bound for every truncation lag and only afterward takes their exact geometric average. The resulting bounds depend on the elapsed lag k, uniformly in the creation time s.

Step 1: auxiliary lag truncations. For $0 \le k < N$, define the analytical lag truncations explicitly by

$$\nabla_\theta \hat{s}_k^1(\theta) = \sum_{n=1}^{N} \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}^J, \qquad \nabla_\theta s_k^1(\theta) = \sum_{n=1}^{N} \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}. \tag{S242}$$

For a fixed creation time s, the terms in the first sum telescope:

$$\begin{aligned} &m(\xi_s)(h_s) + \sum_{n=s+1}^{\min(N,s+k)} \{m(\xi_n)(h_s) - m(\xi_{n-1})(h_s)\} \\ &\quad = m(\xi_{\min(N,s+k)})(h_s). \end{aligned} \tag{S243}$$

Summing Equation (S243) over $s = 1, \ldots, N$ gives

$$\nabla_\theta \hat{s}_k^1(\theta) = \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N)\left(\sum_{s=N-k}^{N} h_s\right), \tag{S244}$$

$$\nabla_\theta s_k^1(\theta) = \sum_{s=1}^{N-k-1} \rho_{s+k}(h_s) + \rho_N\left(\sum_{s=N-k}^{N} h_s\right). \tag{S245}$$

An empty sum in these displays is zero.

At the time it is created, $h_{s,i}$ is a bounded function of the time-s differentiable particle and

$$\operatorname{osc}(h_{s,i}) \le 2G'(\theta). \tag{S246}$$

At time $s+q$, the same quantity is a historical functional of the q subsequent selection–mutation updates. We now localize the historical particle result to this window. Read η_s^J and η_s through their canonical prediction-path lifts, and use the canonical path-space lift of the already defined $Q_{s,s+q}$. Direct addition and subtraction then give

$$\begin{aligned} &m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i}) \\ &\quad = \left\{ m(\xi_{s+q})(h_{s,i}) - \frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)} \right\} \end{aligned}$$

$$+\left\{\frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)} - \frac{\eta_s Q_{s,s+q}(h_{s,i})}{\eta_s Q_{s,s+q}(1)}\right\}. \tag{S247}$$

The first brace is the error made within the window when its initial prediction cloud is held fixed. It contains the final filtering resampling, so first write it as

$$\begin{aligned}
&m(\xi_{s+q})(h_{s,i}) - \frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)} \\
&\quad = \left[m(\xi_{s+q})(h_{s,i}) - \Psi_{s+q}\{m(\xi_{s+q}^P)\}(h_{s,i})\right] \\
&\qquad + \left[\Psi_{s+q}\{m(\xi_{s+q}^P)\}(h_{s,i}) - \Psi_{s+q}\{\Phi_{s,s+q}(\eta_s^J)\}(h_{s,i})\right],
\end{aligned} \tag{S248}$$

where

$$\Psi_{s+q}\{\Phi_{s,s+q}(\eta_s^J)\}(h_{s,i}) = \frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)}. \tag{S249}$$

Conditionally on the prediction paths at time $s+q$, the first bracket in Equation (S248) is a centered multinomial average. Therefore

$$\begin{aligned}
&\mathbb{E}\left[m(\xi_{s+q})(h_{s,i}) - \Psi_{s+q}\{m(\xi_{s+q}^P)\}(h_{s,i}) \,\middle|\, \xi_{s+q}^P\right] = 0, \\
&\mathbb{E}\left[\left|m(\xi_{s+q})(h_{s,i}) - \Psi_{s+q}\{m(\xi_{s+q}^P)\}(h_{s,i})\right|^2 \,\middle|\, \xi_{s+q}^P\right] \le \frac{\operatorname{osc}(h_{s,i})^2}{J}.
\end{aligned} \tag{S250}$$

We next record explicitly the historical prediction-path estimate used for the second bracket. For any bounded f of the path segment from s through $s+q$, put

$$\bar{f} = f - \Phi_{s,s+q}(\eta_s^J)(f). \tag{S251}$$

When $q = 0$, $m(\xi_s^P) = \eta_s^J$, so the prediction error below is identically zero. Suppose therefore that $q \ge 1$. Conditional on η_s^J, the time-$(s+1)$ prediction paths are independent with common law $\Phi_{s,s+1}(\eta_s^J)$; Theorems 3.3 and 3.5 of Chan et al. (2018) may consequently be applied to the remaining $q-1$ selection–mutation updates. Expansion of the empirical square gives

$$\begin{aligned}
&\mathbb{E}\left[\left|m(\xi_{s+q}^P)(f) - \Phi_{s,s+q}(\eta_s^J)(f)\right|^2 \,\middle|\, \eta_s^J\right] \\
&\quad = \frac{1}{J^2}\sum_{j=1}^{J} \mathbb{E}\left\{\bar{f}(\xi_{s+q,j}^P)^2 \,\middle|\, \eta_s^J\right\} \\
&\qquad + \frac{1}{J^2}\sum_{a\neq b} \mathbb{E}\left\{\bar{f}(\xi_{s+q,a}^P)\bar{f}(\xi_{s+q,b}^P) \,\middle|\, \eta_s^J\right\}.
\end{aligned} \tag{S252}$$

The diagonal summands are at most $\operatorname{osc}(f)^2$. For $a \neq b$, choose a constant c such that $0 \le \bar{f} + c \le \operatorname{osc}(f)$. Since the conditional ideal law integrates $\bar{f} + c$ to c, direct expansion gives

$$\begin{aligned}
&\mathbb{E}\left\{\bar{f}(\xi_{s+q,a}^P)\bar{f}(\xi_{s+q,b}^P) \,\middle|\, \eta_s^J\right\} \\
&\quad = \mathbb{E}\left\{\{\bar{f}(\xi_{s+q,a}^P) + c\}\{\bar{f}(\xi_{s+q,b}^P) + c\} \,\middle|\, \eta_s^J\right\} \\
&\qquad - c\mathbb{E}\left\{\bar{f}(\xi_{s+q,a}^P) + c \,\middle|\, \eta_s^J\right\} - c\mathbb{E}\left\{\bar{f}(\xi_{s+q,b}^P) + c \,\middle|\, \eta_s^J\right\} + c^2.
\end{aligned} \tag{S253}$$

The two-lineage case of Theorem 3.5 of Chan et al. (2018) bounds the first expectation minus c^2 by $Cq\,\mathrm{osc}(f)^2/J$, and Theorem 3.3 of the same paper bounds each of the next two expectations minus c by $Cq\,\mathrm{osc}(f)/J$. Since $|c| \le \mathrm{osc}(f)$, substitution into Equation (S252) gives

$$\mathbb{E}\left[\left|m(\xi^P_{s+q})(f) - \Phi_{s,s+q}(\eta^J_s)(f)\right|^2 \,\middle|\, \eta^J_s\right] \le \frac{C(q+1)}{J}\,\mathrm{osc}(f)^2, \tag{S254}$$

$$\left|\mathbb{E}\left[\, m(\xi^P_{s+q})(f) - \Phi_{s,s+q}(\eta^J_s)(f) \,\middle|\, \eta^J_s\right]\right| \le \frac{Cq}{J}\,\mathrm{osc}(f). \tag{S255}$$

Apply these inequalities to the exact Bayes-ratio identity

$$\begin{aligned}&\Psi_{s+q}\{m(\xi^P_{s+q})\}(h_{s,i}) - \Psi_{s+q}\{\Phi_{s,s+q}(\eta^J_s)\}(h_{s,i})\\ &\quad= \frac{\{m(\xi^P_{s+q}) - \Phi_{s,s+q}(\eta^J_s)\}\left[G_{s+q}\left\{h_{s,i} - \Psi_{s+q}\{\Phi_{s,s+q}(\eta^J_s)\}(h_{s,i})\right\}\right]}{m(\xi^P_{s+q})(G_{s+q})}.\end{aligned} \tag{S256}$$

The denominator is at least $\underline{\mathrm{G}}$, and the oscillation of the function in square brackets is at most $2\bar{G}\,\mathrm{osc}(h_{s,i})$. Equation (S254) therefore gives

$$\mathbb{E}\left[\left|\Psi_{s+q}\{m(\xi^P_{s+q})\}(h_{s,i}) - \Psi_{s+q}\{\Phi_{s,s+q}(\eta^J_s)\}(h_{s,i})\right|^2 \,\middle|\, \eta^J_s\right] \le \frac{C(q+1)}{J}\,\mathrm{osc}(h_{s,i})^2. \tag{S257}$$

For the conditional expectation, set, only in the next display,

$$\begin{aligned}&\nu = m(\xi^P_{s+q}), &&\mu = \Phi_{s,s+q}(\eta^J_s),\\ &d = G_{s+q}\{h_{s,i} - \Psi_{s+q}(\mu)(h_{s,i})\}.\end{aligned} \tag{S258}$$

Then $\mu(d) = 0$, and expansion of the reciprocal denominator in Equation (S256) gives the exact identity

$$\frac{(\nu-\mu)(d)}{\nu(G_{s+q})} = \frac{(\nu-\mu)(d)}{\mu(G_{s+q})} - \frac{(\nu-\mu)(d)(\nu-\mu)(G_{s+q})}{\nu(G_{s+q})\mu(G_{s+q})}. \tag{S259}$$

Equation (S255) bounds the conditional expectation of the first term on the right by $Cq\,\mathrm{osc}(h_{s,i})/J$. Conditional Cauchy–Schwarz and Equation (S254) bound the second term by the same order. Hence

$$\left|\mathbb{E}\left[\,\Psi_{s+q}\{m(\xi^P_{s+q})\}(h_{s,i}) - \Psi_{s+q}\{\Phi_{s,s+q}(\eta^J_s)\}(h_{s,i}) \,\middle|\, \eta^J_s\right]\right| \le \frac{Cq}{J}\,\mathrm{osc}(h_{s,i}). \tag{S260}$$

Equations (S248), (S250), (S257), and (S260) prove, conditionally on η^J_s,

$$\mathbb{E}\left[\left|m(\xi_{s+q})(h_{s,i}) - \frac{\eta^J_s Q_{s,s+q}(h_{s,i})}{\eta^J_s Q_{s,s+q}(1)}\right|^2 \,\middle|\, \eta^J_s\right] \le \frac{C(q+1)}{J}\,\mathrm{osc}(h_{s,i})^2, \tag{S261}$$

$$\left|\mathbb{E}\left[m(\xi_{s+q})(h_{s,i}) - \frac{\eta^J_s Q_{s,s+q}(h_{s,i})}{\eta^J_s Q_{s,s+q}(1)} \,\middle|\, \eta^J_s\right]\right| \le \frac{Cq}{J}\,\mathrm{osc}(h_{s,i}). \tag{S262}$$

The second brace in Equation (S247) is the error in the initial prediction cloud. Direct subtraction of its two ratios gives

$$\frac{\eta^J_s Q_{s,s+q}(h_{s,i})}{\eta^J_s Q_{s,s+q}(1)} - \frac{\eta_s Q_{s,s+q}(h_{s,i})}{\eta_s Q_{s,s+q}(1)}$$

$$= \frac{(\eta_s^J - \eta_s) Q_{s,s+q} \left\{ h_{s,i} - \rho_{s+q}(h_{s,i}) \right\}}{\eta_s^J Q_{s,s+q}(1)}. \tag{S263}$$

Because $h_{s,i}$ is created at time s and is preserved along the subsequent path,

$$Q_{s,s+q}\{h_{s,i} - \rho_{s+q}(h_{s,i})\} = \{h_{s,i} - \rho_{s+q}(h_{s,i})\} Q_{s,s+q}(1). \tag{S264}$$

Equation (S201) therefore gives

$$\begin{aligned} \inf_x \frac{Q_{s,s+q}(1)(x)}{\eta_s Q_{s,s+q}(1)} &\geq C^{-1}, \\ \operatorname{osc}\left[\frac{Q_{s,s+q}\{h_{s,i} - \rho_{s+q}(h_{s,i})\}}{\eta_s Q_{s,s+q}(1)} \right] &\leq C \operatorname{osc}(h_{s,i}). \end{aligned} \tag{S265}$$

Lemma 2.3 of Karjalainen et al. (2023), which concerns the prediction empirical measure, and Equation (S265) therefore give

$$\mathbb{E} \left| \frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)} - \frac{\eta_s Q_{s,s+q}(h_{s,i})}{\eta_s Q_{s,s+q}(1)} \right|^2 \leq \frac{C}{J} \operatorname{osc}(h_{s,i})^2. \tag{S266}$$

For its expectation, use the exact ratio cancellation in Equations (S216)– (S217) and the update telescope in Equation (S218). The same bounds give

$$\left| \mathbb{E} \left[\frac{\eta_s^J Q_{s,s+q}(h_{s,i})}{\eta_s^J Q_{s,s+q}(1)} - \frac{\eta_s Q_{s,s+q}(h_{s,i})}{\eta_s Q_{s,s+q}(1)} \right] \right| \leq \frac{C}{J} \operatorname{osc}(h_{s,i}). \tag{S267}$$

Apply $|a+b|^2 \leq 2a^2 + 2b^2$ to Equation (S247), use Equations (S261) and (S266), and then use Equation (S246). For $J \geq c(q+1)$ this gives

$$\mathbb{E} \left| m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i}) \right|^2 \leq \frac{C(q+1) G'(\theta)^2}{J}. \tag{S268}$$

Taking expectations in Equation (S247) and combining Equations (S262) and (S267) gives

$$\left| \mathbb{E} m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i}) \right| \leq \frac{C(q+1) G'(\theta)}{J}. \tag{S269}$$

The particle-number restriction in the cited theorems does not restrict these conclusions. If $2 \leq J < c(q+1)$, then $(q+1)/J > 1/c$, while the pathwise differences are at most $2G'(\theta)$. After increasing C, Equations (S268) and (S269) therefore hold for every $J \geq 2$, $q \in \{0, \dots, k\}$, and $1 \leq s \leq N - q$.

The terminal term is one additive historical functional over $k+1$ times, not a current-particle function. Its coordinatewise oscillation satisfies

$$\operatorname{osc}\left(\sum_{s=N-k}^{N} h_{s,i} \right) \leq 2(k+1) G'(\theta). \tag{S270}$$

The exact analogue of Equation (S247) is

$$m(\xi_N) \left(\sum_{s=N-k}^{N} h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^{N} h_{s,i} \right)$$

$$= \left\{ m(\xi_N) \left(\sum_{s=N-k}^{N} h_{s,i} \right) - \frac{\eta^J_{N-k} Q_{N-k,N} \left(\sum_{s=N-k}^{N} h_{s,i} \right)}{\eta^J_{N-k} Q_{N-k,N}(1)} \right\}$$

$$+ \left\{ \frac{\eta^J_{N-k} Q_{N-k,N} \left(\sum_{s=N-k}^{N} h_{s,i} \right)}{\eta^J_{N-k} Q_{N-k,N}(1)} - \frac{\eta_{N-k} Q_{N-k,N} \left(\sum_{s=N-k}^{N} h_{s,i} \right)}{\eta_{N-k} Q_{N-k,N}(1)} \right\}. \tag{S271}$$

The first brace is controlled by the diagonal–off-diagonal calculation in Equation (S261) over $k+1$ times. For the second brace, the normalized conditional expectation of the displayed sum lies between its infimum and supremum. The ratio calculation in Equations (S263)– (S267) therefore applies with the oscillation in Equation (S270). Thus

$$\mathbb{E} \left| m(\xi_N) \left(\sum_{s=N-k}^{N} h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^{N} h_{s,i} \right) \right|^2 \leq \frac{C(k+1)^3 G'(\theta)^2}{J}, \tag{S272}$$

$$\left| \mathbb{E} m(\xi_N) \left(\sum_{s=N-k}^{N} h_{s,i} \right) - \rho_N \left(\sum_{s=N-k}^{N} h_{s,i} \right) \right| \leq \frac{C(k+1)^2 G'(\theta)}{J}. \tag{S273}$$

For $J < c(k+1)$ these inequalities again follow from the pathwise bounds after increasing C.

We next sum the covariances in the first term of Equation (S244). Cauchy–Schwarz and Equation (S268) give, for $s < t < N-k$,

$$\begin{aligned} &|\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}), m(\xi_{t+k})(h_{t,i})\}| \\ &\quad \leq [\mathrm{Var}\{m(\xi_{s+k})(h_{s,i})\}\mathrm{Var}\{m(\xi_{t+k})(h_{t,i})\}]^{1/2} \\ &\quad \leq \frac{C(k+1)G'(\theta)^2}{J}. \end{aligned} \tag{S274}$$

If $t - s > k$, the first quantity is measurable through time $s+k$. The second is obtained from the time-t differentiable particle and the subsequent selection–mutation variables through time $t+k$. The two windows are separated by $t-s-k$ particle updates. Davydov's inequality, the coefficient $a(\cdot)$ in Equation (S135), and the pathwise bound give

$$\|m(\xi_{s+k})(h_{s,i}) - \mathbb{E}m(\xi_{s+k})(h_{s,i})\|_{L^4} \leq 2G'(\theta), \tag{S275}$$

and the same inequality with s replaced by t. Covariance is unchanged by this centering, so Davydov's inequality gives

$$|\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}), m(\xi_{t+k})(h_{t,i})\}| \leq CG'(\theta)^2 a(t-s-k)^{1/2}. \tag{S276}$$

Combining Equations (S274) and (S276), and separating the k overlapping separations, gives

$$\begin{aligned} &\sum_{t=s+1}^{N-k-1} |\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}), m(\xi_{t+k})(h_{t,i})\}| \\ &\quad = \sum_{d=1}^{N-k-1-s} |\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}), m(\xi_{s+d+k})(h_{s+d,i})\}| \end{aligned}$$

$$
\begin{aligned}
&\le \frac{Ck(k+1)G'(\theta)^2}{J} + CG'(\theta)^2 \sum_{d=k+1}^{N-k-1-s} \min\left\{\frac{k+1}{J}, a(d-k)^{1/2}\right\} \\
&\lesssim \frac{C(k+1)^2G'(\theta)^2}{J}.
\end{aligned} \tag{S277}
$$

The inequality holds for every $1 \le s \le N-k-2$; an empty sum is zero. The last inequality uses the geometric bound on $a(\cdot)$ in Equation (S135). More explicitly, that bound gives

$$
\sum_{u=1}^{\infty} \min\left\{\frac{k+1}{J}, a(u)^{1/2}\right\} \le \frac{C(k+1)}{J}(1+\log J)^2. \tag{S278}
$$

As in the preceding variance arguments, $\lesssim$ suppresses this particle-system mixing and logarithmic-threshold factor. Expanding the variance and using Equation (S277) now gives

$$
\begin{aligned}
&\operatorname{Var}\left\{\sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_{s,i})\right\} \\
&= \sum_{s=1}^{N-k-1} \operatorname{Var}\{m(\xi_{s+k})(h_{s,i})\} \\
&\quad + 2\sum_{s=1}^{N-k-2}\sum_{t=s+1}^{N-k-1} \operatorname{Cov}\{m(\xi_{s+k})(h_{s,i}), m(\xi_{t+k})(h_{t,i})\} \\
&\lesssim \frac{N(k+1)^2G'(\theta)^2}{J}.
\end{aligned} \tag{S279}
$$

Equation (S272) and $k+1 \le N$ give

$$
\begin{aligned}
\operatorname{Var}\left\{m(\xi_N)\left(\sum_{s=N-k}^{N} h_{s,i}\right)\right\} &\le \mathbb{E}\left|m(\xi_N)\left(\sum_{s=N-k}^{N} h_{s,i}\right) - \rho_N\left(\sum_{s=N-k}^{N} h_{s,i}\right)\right|^2 \\
&\le \frac{C(k+1)^3G'(\theta)^2}{J} \\
&\le \frac{CN(k+1)^2G'(\theta)^2}{J}.
\end{aligned} \tag{S280}
$$

Writing the two terms in Equation (S244) as A and B, respectively, and using

$$
\operatorname{Var}(A+B) \le 2\operatorname{Var}(A) + 2\operatorname{Var}(B), \tag{S281}
$$

then summing over the p coordinates, proves

$$
\operatorname{Var}\{\nabla_\theta \hat{s}_k^1(\theta)\} \lesssim \frac{NpG'(\theta)^2(k+1)^2}{J}. \tag{S282}
$$

Equations (S269) and (S273), the triangle inequality, and $k+1 \le N$ similarly give

$$
\left\|\mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 \le \frac{C\sqrt{p}\,G'(\theta)}{J}\left\{(N-k-1)(k+1) + (k+1)^2\right\}
$$

$$\leq \frac{CN\sqrt{p}\,G'(\theta)(k+1)}{J}. \tag{S283}$$

Step 2: recover geometric discounting exactly. For $0 \leq k < N-1$, put

$$\beta_k = (1-\alpha)\alpha^k, \qquad \beta_{N-1} = \alpha^{N-1}. \tag{S284}$$

These nonnegative numbers sum to one. Moreover, for every $0 \leq r < N$,

$$\begin{aligned} \sum_{k=r}^{N-1} \beta_k &= \sum_{k=r}^{N-2} (1-\alpha)\alpha^k + \alpha^{N-1} \\ &= \alpha^r(1-\alpha^{N-1-r}) + \alpha^{N-1} \\ &= \alpha^r. \end{aligned} \tag{S285}$$

Equations (S157), (S161), and (S242) therefore give the pathwise identities

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_\theta \hat{s}_k^1(\theta), \qquad \nabla_\theta \ell^\alpha(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_\theta s_k^1(\theta). \tag{S286}$$

Indeed, the coefficient of $\Delta_{n,r}^J$, or of $\Delta_{n,r}$, on the right is exactly the tail sum in Equation (S285). Thus this is an identity for the implemented geometrically discounted estimator, not a change to the algorithm and not an additional randomization. In this precise sense, geometric discounting is a soft lag truncation whose effective memory is computed below.

There are two different coefficients at lag zero. The number $\beta_0 = 1-\alpha$ is the mixture weight on the entire hard-zero-lag score $\nabla_\theta \hat{s}_0^1(\theta)$. By contrast, every hard-k score contains $\Delta_{n,0}^J$, so the coefficient of the zero-lag contribution in the geometric mixture is

$$\sum_{k=0}^{N-1} \beta_k = 1. \tag{S287}$$

In particular, at $\alpha = 0$,

$$\nabla_\theta \hat{\ell}_n^0(\theta) = \Delta_{n,0}^J, \qquad \nabla_\theta \ell_n^0(\theta) = \Delta_{n,0}. \tag{S288}$$

The mean lag in this censored geometric mixture satisfies

$$\begin{aligned} \sum_{k=0}^{N-1} \beta_k (k+1) &= \sum_{r=0}^{N-1} \sum_{k=r}^{N-1} \beta_k \\ &= \sum_{r=0}^{N-1} \alpha^r. \end{aligned} \tag{S289}$$

Center the first identity in Equation (S286). Minkowski's inequality, Equation (S282), and Equation (S289) give

$$\left\| \nabla_\theta \hat{\ell}^\alpha(\theta) - \mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) \right\|_{L^2} \leq \sum_{k=0}^{N-1} \beta_k \left\| \nabla_\theta \hat{s}_k^1(\theta) - \mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) \right\|_{L^2}$$

$$
\begin{aligned}
&\lesssim C\sqrt{\frac{NpG'(\theta)^2}{J}}\sum_{k=0}^{N-1}\beta_k(k+1)\\
&\lesssim C\sqrt{\frac{NpG'(\theta)^2}{J}}\sum_{r=0}^{N-1}\alpha^r. \qquad \text{(S290)}
\end{aligned}
$$

Squaring proves Equation (S238). Taking expectations in Equation (S286), applying the triangle inequality and Equation (S283), and then using Equation (S289) give

$$
\begin{aligned}
\left\|\mathbb{E}\nabla_\theta\hat{\ell}^\alpha(\theta)-\nabla_\theta\ell^\alpha(\theta)\right\|_2 &\le \sum_{k=0}^{N-1}\beta_k\left\|\mathbb{E}\nabla_\theta\hat{s}_k^1(\theta)-\nabla_\theta s_k^1(\theta)\right\|_2\\
&\le \frac{CN\sqrt{p}\,G'(\theta)}{J}\sum_{k=0}^{N-1}\beta_k(k+1)\\
&= \frac{CN\sqrt{p}\,G'(\theta)}{J}\sum_{r=0}^{N-1}\alpha^r, \qquad \text{(S291)}
\end{aligned}
$$

which proves Equation (S239). Finally, for a random vector Z and deterministic z,

$$
\mathbb{E}\|Z-z\|_2^2=\mathrm{Var}(Z)+\|\mathbb{E}Z-z\|_2^2. \qquad \text{(S292)}
$$

Apply this identity with $Z=\nabla_\theta\hat{\ell}^\alpha(\theta)$ and $z=\nabla_\theta\ell^\alpha(\theta)$, and substitute Equations (S238) and (S239). This proves Equation (S240). □

Corollary S1 (Particle bias of the DMOP-α score)**.** *Under the conditions of Lemma S13, Equation (S239) holds.*

Proof. This is the second conclusion of Lemma S13. □

Lemma S14 (Population discounting bias)**.** *Under the conditions of Lemma S10, suppose $\theta=\phi$ and $\alpha\in[0,1]$. Then*

$$
\begin{aligned}
\|\nabla_\theta\ell_n^\alpha(\theta)-\nabla_\theta\ell_n(\theta)\|_2 &\le C\sqrt{p}\,G'(\theta)\sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r\\
&\le C\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \qquad \text{(S293)}
\end{aligned}
$$

For the full score,

$$
\begin{aligned}
&\|\nabla_\theta\ell(\theta)-\nabla_\theta\ell^\alpha(\theta)\|_2\\
&\quad\le C\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q\\
&\quad= C\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\frac{(N-1)-N\alpha\varrho+(\alpha\varrho)^N}{(1-\alpha\varrho)^2}\\
&\quad\le C(N-1)\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}. \qquad \text{(S294)}
\end{aligned}
$$

For $N = 1$, the finite sum is empty, the displayed fraction is zero, and the full-score difference is zero.

Proof. Step 1: exact lagwise difference. Equation (S161) at α and at $\alpha = 1$ gives

$$\begin{aligned}\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1}(1-\alpha^r)\Delta_{n,r} \\ &= \sum_{r=1}^{n-1}(1-\alpha^r)\Delta_{n,r}, \end{aligned} \tag{S295}$$

The coefficient of $\Delta_{n,0}$ is one in both scores, so its difference is zero. In particular, at the left endpoint Equation (S288) gives

$$\nabla_\theta \ell_n^0(\theta) = \Delta_{n,0}, \qquad \nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^0(\theta) = \sum_{r=1}^{n-1} \Delta_{n,r}. \tag{S296}$$

Thus DMOP-0 retains the zero-lag contribution with coefficient one and retains no positive-lag contribution.

Step 2: apply forgetting and sum. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}\|\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\|_2 &\le \sum_{r=1}^{n-1}(1-\alpha^r)\|\Delta_{n,r}\|_2 \\ &\le C\sqrt{p}\, G'(\theta) \sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r \\ &\le C\sqrt{p}\, G'(\theta) \sum_{r=1}^{\infty}(1-\alpha^r)\varrho^r \\ &= C\sqrt{p}\, G'(\theta) \left\{\sum_{r=1}^{\infty}\varrho^r - \sum_{r=1}^{\infty}(\alpha\varrho)^r\right\} \\ &= C\sqrt{p}\, G'(\theta) \left\{\frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho}\right\} \\ &= C\sqrt{p}\, G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \end{aligned} \tag{S297}$$

Step 3: exact summation by parts for the full score. Reindexing Equation (S161) by $s = n - r$ shows that the contribution of h_s to the full DMOP-α population score is

$$\begin{aligned}&\rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q\{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\ &= \rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \rho_{s+q}(h_s) - \sum_{q=0}^{N-s-1} \alpha^{q+1}\rho_{s+q}(h_s)\end{aligned}$$

$$
= (1-\alpha)\sum_{q=0}^{N-s-1}\alpha^q\rho_{s+q}(h_s) + \alpha^{N-s}\rho_N(h_s). \tag{S298}
$$

The contribution of h_s to the exact score in Equation (S159) is $\rho_N(h_s)$. Subtracting Equation (S298) from this contribution gives

$$
\begin{aligned}
&\rho_N(h_s) - \rho_s(h_s) - \sum_{q=1}^{N-s}\alpha^q\{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&= (1-\alpha)\sum_{q=0}^{N-s-1}\alpha^q\{\rho_N(h_s) - \rho_{s+q}(h_s)\}.
\end{aligned} \tag{S299}
$$

Summing Equation (S299) over $s = 1, \dots, N-1$ yields the exact identity

$$
\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) = (1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}\alpha^q\{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S300}
$$

Step 4: contract each partial-smoother comparison. Fix $s \le N-1$, $0 \le q \le N-s-1$, and a coordinate $i \in \{1, \dots, p\}$. The Markov property implies that observations after time $s+q$ are conditionally independent of the path through time s given X_{s+q}. Hence the following two conditional expectations are the same function of x:

$$
\begin{aligned}
\mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\} &= \mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q} = x\} \\
&= \begin{cases} \bar{h}_{s,i}(x), & q = 0, \\ (B_{s+q}B_{s+q-1}\cdots B_{s+1}\bar{h}_{s,i})(x), & q \ge 1. \end{cases}
\end{aligned} \tag{S301}
$$

Here $\bar{h}_{s,i}$ is the conditional projection in Equation (S174); its invariance under later posterior laws is Equation (S177). Also, B_t is the posterior backward kernel in Equation (S168). Equation (S179) and q successive applications of Equation (S173) give

$$
\begin{aligned}
&\operatorname{osc}[x \longmapsto \mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\}] \\
&\quad \le 2G'(\theta)(1-\epsilon)^q \le 2G'(\theta)\varrho^q.
\end{aligned} \tag{S302}
$$

The observations after time $s+q$ supply one additional contraction. Since $s+q < N$, Equation (S175) and the Markov property give

$$
L_{s+q,N}(x) = \int M_{s+q+1}(x, du)G_{s+q+1}(u)L_{s+q+1,N}(u). \tag{S303}
$$

For every x, x', Assumption (A2) gives

$$
\begin{aligned}
L_{s+q,N}(x) &\ge \epsilon \int M_{s+q+1}(x', du)G_{s+q+1}(u)L_{s+q+1,N}(u) \\
&= \epsilon L_{s+q,N}(x').
\end{aligned} \tag{S304}
$$

Consequently,

$$\frac{L_{s+q,N}(x)}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}} \geq \epsilon. \tag{S305}$$

Indeed, the denominator is no larger than $\sup_{x'} L_{s+q,N}(x')$, while Equation (S304) bounds the numerator below by $\epsilon \sup_{x'} L_{s+q,N}(x')$.

For $\epsilon < 1$, subtracting ϵ from the left side of Equation (S305) and dividing by $1-\epsilon$ produces a nonnegative function whose ρ_{s+q} expectation is one. Hence, for every bounded real function f of X_{s+q},

$$\begin{aligned}
&\frac{\rho_{s+q}\{f(X_{s+q})L_{s+q,N}(X_{s+q})\}}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}} \\
&\quad = \epsilon\rho_{s+q}\{f(X_{s+q})\} \\
&\qquad + (1-\epsilon)\rho_{s+q}\left[f(X_{s+q})\frac{L_{s+q,N}(X_{s+q})/\rho_{s+q}\{L_{s+q,N}(X_{s+q})\} - \epsilon}{1-\epsilon}\right].
\end{aligned} \tag{S306}$$

Subtracting $\rho_{s+q}\{f(X_{s+q})\}$ from both sides of Equation (S306) and using that both terms multiplied by ϵ and $1-\epsilon$ are probability expectations gives

$$\begin{aligned}
&\left|\frac{\rho_{s+q}\{f(X_{s+q})L_{s+q,N}(X_{s+q})\}}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}} - \rho_{s+q}\{f(X_{s+q})\}\right| \\
&\quad \leq (1-\epsilon)\operatorname{osc}(f).
\end{aligned} \tag{S307}$$

If $\epsilon = 1$, the two expectations in Equation (S307) are equal, so the same inequality holds.

Equation (S150), conditioning on X_{s+q}, and Equation (S301) give

$$\begin{aligned}
\rho_N(h_{s,i}) &= \frac{\rho_{s+q}\left[\mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q}\}L_{s+q,N}(X_{s+q})\right]}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}}, \\
\rho_{s+q}(h_{s,i}) &= \rho_{s+q}\left[\mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q}\}\right].
\end{aligned} \tag{S308}$$

Apply Equation (S307) to the common conditional expectation in Equation (S301). The tower identities in Equation (S308) and Equation (S302) give

$$\begin{aligned}
|\rho_N(h_{s,i}) - \rho_{s+q}(h_{s,i})| &\leq 2G'(\theta)(1-\epsilon)^{q+1} \\
&\leq 2G'(\theta)\varrho^{q+1}.
\end{aligned} \tag{S309}$$

Summing the squared coordinate bounds and taking square roots yield

$$\|\rho_N(h_s) - \rho_{s+q}(h_s)\|_2 \leq 2\sqrt{p}\, G'(\theta)\varrho^{q+1}. \tag{S310}$$

Insert Equation (S310) into Equation (S300), apply the triangle inequality, and reverse the two finite sums. This gives

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad \leq 2\sqrt{p}\, G'(\theta)\varrho(1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}(\alpha\varrho)^q
\end{aligned}$$

$$
\begin{aligned}
&= 2\sqrt{p}\, G'(\theta)\varrho(1-\alpha) \sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q \\
&\leq 2(N-1)\sqrt{p}\, G'(\theta)\varrho(1-\alpha) \sum_{q=0}^{\infty} (\alpha\varrho)^q \\
&= 2(N-1)\sqrt{p}\, G'(\theta) \frac{\varrho(1-\alpha)}{1-\alpha\varrho}. && \text{(S311)}
\end{aligned}
$$

Finally, multiplying the finite sum by $1-\alpha\varrho$ gives

$$
\begin{aligned}
&(1-\alpha\varrho) \sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q \\
&\qquad = (N-1) - \sum_{q=1}^{N-1} (\alpha\varrho)^q \\
&\qquad = (N-1) - \frac{\alpha\varrho\{1-(\alpha\varrho)^{N-1}\}}{1-\alpha\varrho}. && \text{(S312)}
\end{aligned}
$$

Multiplying Equation (S312) once more by $1-\alpha\varrho$ and collecting terms therefore gives

$$
\sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q = \frac{(N-1) - N\alpha\varrho + (\alpha\varrho)^N}{(1-\alpha\varrho)^2}. \tag{S313}
$$

Equations (S311) and (S313) prove Equation (S294). □

Remark 1 (Sharpness of the fixed-observation population order)**.** *The powers of N and $1-\alpha$ in Equation (S294) cannot be improved uniformly under the conditions of Lemma S14. To see this, take state space $\{0,1\}$ with counting measure,*

$$
X_0 \sim \text{Bernoulli}(3/4), \qquad \Pr(X_n = X_{n-1} \mid X_{n-1}) = \frac{3}{4}, \tag{S314}
$$

fix $y_n^* = 1$ for every n, and, for $|\theta| \leq 1/4$, set

$$
g_\theta(1 \mid 0) = \frac{1}{4}, \qquad g_\theta(1 \mid 1) = \frac{9}{20}(1+\theta), \tag{S315}
$$

with the probabilities at observation zero defined by complementation. At $\theta = \phi = 0$,

$$
h_n(x) = x. \tag{S316}
$$

If the filtering probability of state one at time $n-1$ is $3/4$, its one-step prediction and update are

$$
\begin{aligned}
\Pr(X_n = 1 \mid y_{1:n-1}^*) &= \frac{3}{4}\frac{3}{4} + \frac{1}{4}\frac{1}{4} = \frac{5}{8}, \\
\Pr(X_n = 1 \mid y_{1:n}^*) &= \frac{(5/8)(9/20)}{(5/8)(9/20) + (3/8)(1/4)} = \frac{3}{4}. && \text{(S317)}
\end{aligned}
$$

The law of X_0 initializes this induction, so the filtered probability is $3/4$ at every time. The resulting time-homogeneous posterior backward kernel satisfies

$$B(1 \mid 1) = \frac{(3/4)(3/4)}{5/8} = \frac{9}{10}, \qquad B(1 \mid 0) = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}. \tag{S318}$$

Moreover, the next-observation likelihoods conditional on X_{n-1} are

$$M_n(G_n)(1) = \frac{3}{4}\frac{9}{20} + \frac{1}{4}\frac{1}{4} = \frac{2}{5}, \qquad M_n(G_n)(0) = \frac{1}{4}\frac{9}{20} + \frac{3}{4}\frac{1}{4} = \frac{3}{10}. \tag{S319}$$

Consequently,

$$\begin{aligned}\Pr_{\rho_n}(X_{n-1} = 1) &= \frac{(3/4)(2/5)}{(3/4)(2/5) + (1/4)(3/10)} = \frac{4}{5},\\ \Pr_{\rho_{n-1}}(X_{n-1} = 1) &= \frac{3}{4}.\end{aligned} \tag{S320}$$

Equation (S318) maps every affine function of the state to an affine function whose slope is multiplied by

$$B(1 \mid 1) - B(1 \mid 0) = \frac{2}{5}. \tag{S321}$$

Applying the $r-1$ backward kernels between X_{n-r} and X_{n-1} and then using Equation (S320) therefore gives, for $1 \le r < n$,

$$\begin{aligned}\Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\\ &= \left(\frac{4}{5} - \frac{3}{4}\right)\left(\frac{2}{5}\right)^{r-1} = \frac{1}{20}\left(\frac{2}{5}\right)^{r-1}.\end{aligned} \tag{S322}$$

All these quantities are nonnegative. Equation (S295), summed over n, now implies

$$\begin{aligned}\nabla_\theta \ell(0) - \nabla_\theta \ell^\alpha(0) &= \frac{1}{20}\sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\left(\frac{2}{5}\right)^{r-1}\\ &\ge \frac{(N-1)(1-\alpha)}{20}.\end{aligned} \tag{S323}$$

Thus the squared population bias is at least $(N-1)^2(1-\alpha)^2/400$. This model satisfies the uniform transition and measurement bounds on $|\theta| \le 1/4$, its simulator is independent of θ, and h_n is a bounded current function. Improving either power uniformly would therefore require an additional signed-cancellation condition, not merely the existing mixing assumptions.

Theorem S5 (MSE bounds for DMOP-α)**.** *Under the conditions of Lemmas S13 and S14, for $\alpha \in [0,1]$,*

$$\begin{aligned}&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2\\ &\lesssim pG'(\theta)^2\left[\left(\frac{N}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2 + \left\{\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q\right\}^2\right].\end{aligned} \tag{S324}$$

For $0 \le \alpha < 1$, this finite bound implies

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} + N^2\left\{\frac{\varrho(1-\alpha)}{1-\alpha\varrho}\right\}^2\right]. \tag{S325}$$

Proof. Add and subtract $\nabla_\theta \ell^\alpha(\theta)$ and use $\|a+b\|_2^2 \le 2\|a\|_2^2 + 2\|b\|_2^2$. This gives

$$\begin{aligned}&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\ &\quad\le 2\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 + 2\left\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2.\end{aligned} \tag{S326}$$

Lemma S13 bounds the first term by

$$CpG'(\theta)^2\left(\frac{N}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{S327}$$

Lemma S14 gives

$$\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \le CpG'(\theta)^2\left\{\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q\right\}^2. \tag{S328}$$

Substitution into Equation (S326) proves the first bound. For $0 \le \alpha < 1$,

$$\begin{aligned}\sum_{r=0}^{N-1}\alpha^r &\le \frac{1}{1-\alpha}, \\ \varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q &\le N\frac{\varrho(1-\alpha)}{1-\alpha\varrho}.\end{aligned} \tag{S329}$$

Substitution of these two inequalities proves Equation (S325). □

Remark 2 (Neutral-model check)**.** *The following calculation checks the $\alpha = 0$ endpoint and the fixed-$\alpha < 1$ orders in Lemma S13. Take iid* Uniform$[0,1]$ *states, constant weights at $\theta = \phi = 0$, and*

$$\Pr_\theta(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x-1), \qquad y_n^* = 1. \tag{S330}$$

Then $h_n = (2X_n - 1)/2$ has variance $\sigma^2 = 1/12$, and direct conditional multinomial calculation gives

$$\mathrm{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} = \frac{\sigma^2}{J}\left[N(2-J^{-1}) + \sum_{r=1}^{N-1}(N-r)\alpha^{2r}(1-J^{-1})^{r+1}\right]. \tag{S331}$$

Thus the variance has order N/J for every fixed $\alpha < 1$, and its genealogical contribution has order $N/\{J(1-\alpha^2)\}$ before horizon saturation. At $\alpha = 0$, this has order N/J. This calculation also shows that the factor $(1-\alpha)^{-2}$ in the general upper bound is conservative in this model; the theorem does not claim that dependence on α is sharp.

Remark 3 (What the general bound establishes about α)**.** *The finite particle term in Lemma S13 is*

$$pG'(\theta)^2\left(\frac{N}{J}+\frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{S332}$$

Thus $\alpha=0$ gives the DMOP-0 particle order $pG'(\theta)^2(N/J+N^2/J^2)$, and every fixed $\alpha<1$ has the same N/J leading order when $J\geq N$, multiplied by a constant depending on α. The deterioration occurs through the effective memory $\sum_{r=0}^{N-1}\alpha^r$ as $\alpha\uparrow 1$. This is precisely the forgetting induced by discounting: a contribution of age r is multiplied by α^r before the lag terms are summed.

The population term moves in the opposite direction. The finite bound in Theorem S5 decreases to zero as $\alpha\uparrow 1$. Hence the upper bound exhibits the intended bias–variance tradeoff. The neutral calculation in Equation (S331) shows that the theorem's $(1-\alpha)^{-2}$ dependence is conservative in some models.

Oscillation normalization is a separate matter. A particle MSE bound proved for $\operatorname{osc}(f)\leq 1$ *is multiplied by* $\operatorname{osc}(f)^2$ *when it is applied to an unnormalized functional. Thus a generic path-space bound of order N^2/J for an oscillation-one functional becomes N^4/J if it is applied once to a full score whose oscillation is of order N. Lemma S13 does not make that application: it first controls each h_s, whose oscillation is bounded independently of N, and only then performs the discounted lag sum. The N^4/J calculation is therefore a generic norm-rescaling upper bound, not an additional factor to insert into Theorem S5 and not the quadratic asymptotic-variance lower bound of Poyiadjis et al. (2011).*

Remark 4 (Choosing α in the upper bound)**.** *This calculation concerns the proved upper bound, not the unknown exact MSE. Since*

$$\sum_{r=0}^{N-1}\alpha^r\leq\min\{N,(1-\alpha)^{-1}\} \tag{S333}$$

and the fixed model constants include $\varrho/(1-\varrho)$, Theorem S5 gives

$$\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta)-\nabla_\theta\ell(\theta)\right\|_2^2\lesssim pG'(\theta)^2\left[\left(\frac{N}{J}+\frac{N^2}{J^2}\right)\min\{N^2,(1-\alpha)^{-2}\}+N^2(1-\alpha)^2\right]. \tag{S334}$$

Before the effective memory reaches the horizon, balancing the two terms in Equation (S334) gives the choices and attained upper bounds

regime	$1-\alpha$	*MSE upper bound*$/\{pG'(\theta)^2\}$
$J\lesssim N$	$J^{-1/2}$	N^2/J
$N\lesssim J\lesssim N^3$	$(NJ)^{-1/4}$	$N^{3/2}/J^{1/2}$
$J\gtrsim N^3$	0	$N^3/J.$

(S335)

In the last regime, $\alpha=1$ uses the finite sum in Equation (S241); more generally, $1-\alpha\lesssim\sqrt{N/J}$ attains the same order. The first-regime $J^{-1/2}$ behavior reflects the conservative squared effective-memory bound in Lemma S13. The neutral-model calculation shows that this dependence on α need not be sharp. The three rows use the manuscript's $\lesssim$ convention and therefore suppress the same particle-system logarithmic factor as Lemma S13.

Remark 5 (Geometric discounting and lag truncation)**.** *The lag-truncated expressions used in Lemma S13 are analytical devices, not alternative implementations of DMOP-α. For $0 \le k < N$, the truncated score and its population truncation error are*

$$\nabla_\theta \hat{s}_k^1(\theta) = \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N)\left(\sum_{s=N-k}^{N} h_s\right),$$

$$\nabla_\theta \ell(\theta) - \nabla_\theta s_k^1(\theta) = \sum_{s=1}^{N-k-1} \sum_{u=s+k+1}^{N} \Delta_{u,u-s}. \tag{S336}$$

Equation (S310) with $q = k$ gives

$$\left\|\nabla_\theta \ell(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 \le C(N-k-1)\sqrt{p}\, G'(\theta)\varrho^{k+1}. \tag{S337}$$

The finite-window particle calculation in Equations (S282) and (S283) gives

$$\operatorname{Var}\{\nabla_\theta \hat{s}_k^1(\theta)\} \lesssim \frac{NpG'(\theta)^2(k+1)^2}{J},$$

$$\left\|\mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 \lesssim \frac{N\sqrt{p}\, G'(\theta)(k+1)}{J}. \tag{S338}$$

Geometric discounting recovers the implemented estimator exactly by Equation (S286); no cutoff is selected by the algorithm. Thus DMOP-α is exactly a geometric average of the analytical lag truncations and, in this precise sense, approximates a hard lag cutoff. The identity

$$\sum_{k=0}^{N-1} \beta_k(k+1) = \sum_{r=0}^{N-1} \alpha^r \tag{S339}$$

shows why every fixed $\alpha < 1$ has an N/J leading particle-variance order and why $\alpha = 0$ reduces exactly to the DMOP-0 order.

S8 Figures for Bayesian Inference

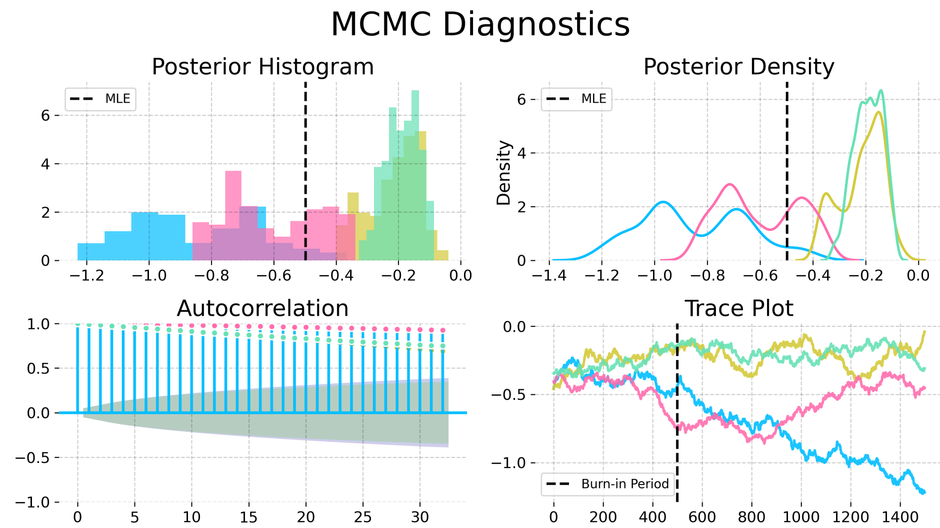

Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

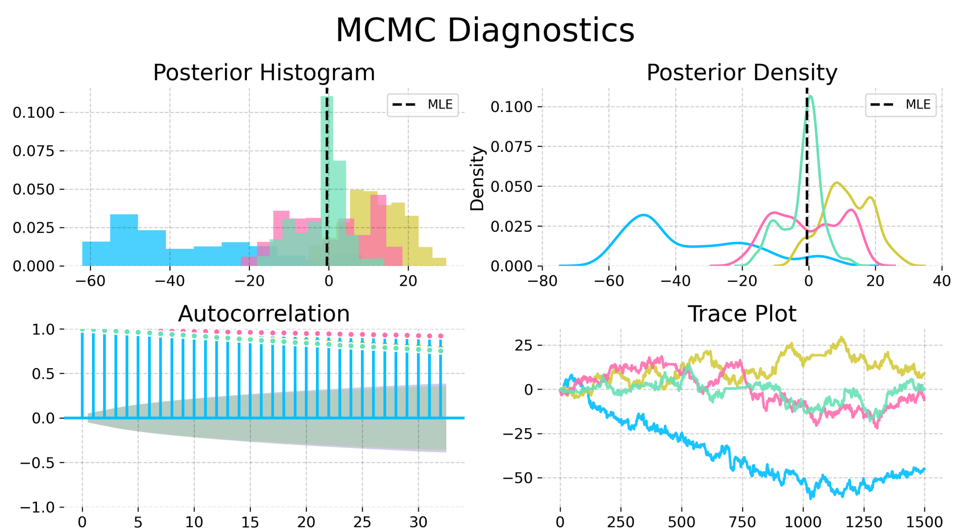

Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

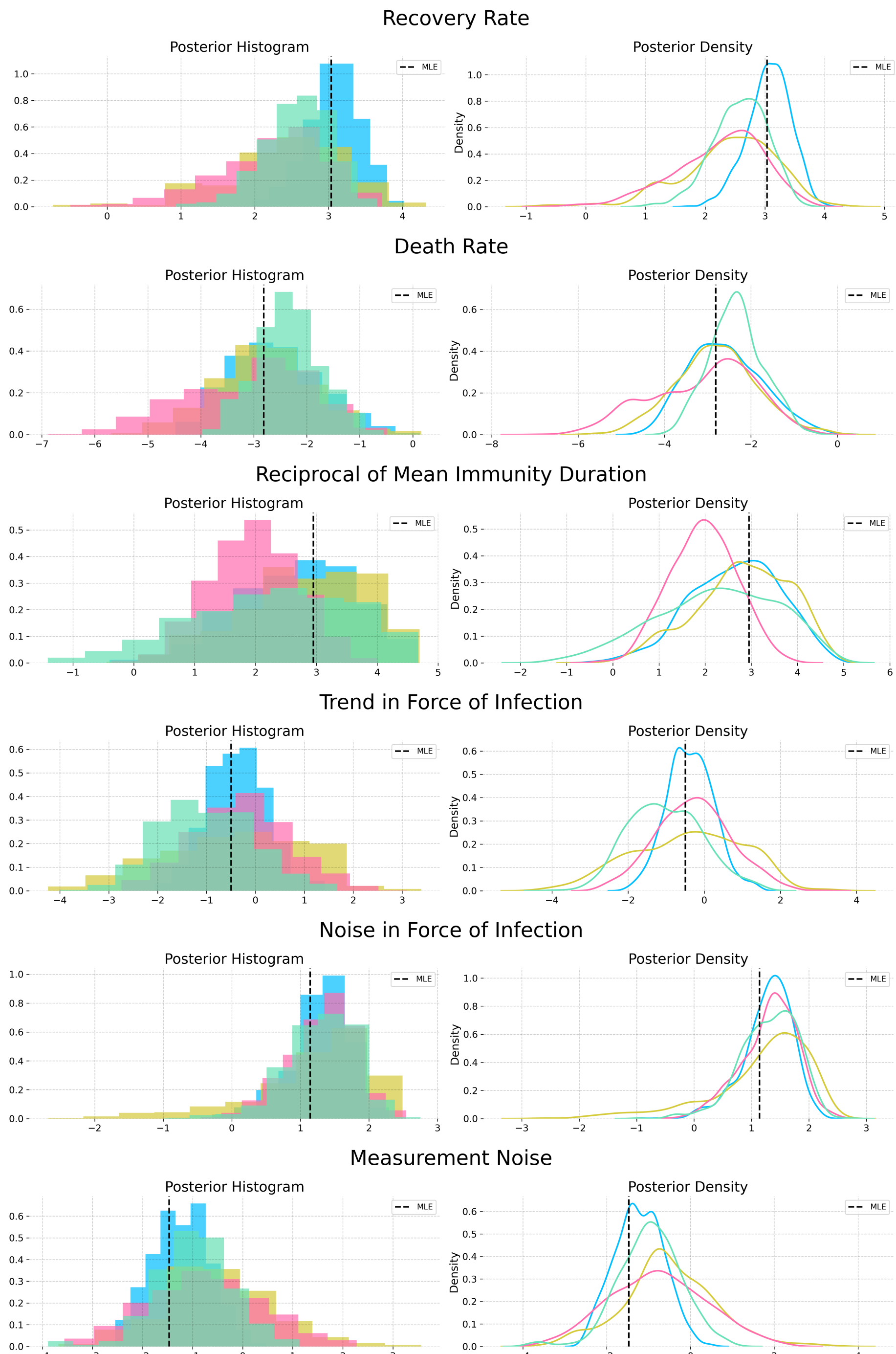

Figure S-3: Posterior estimates from NUTS powered by MOP-α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016) , 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0, are treated differently: they are perturbed only at time t_0, with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	**Standard IF2**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	650	60	175	60
GD Iterations	—	225	175	225
IF2 Cooling Rate	0.8	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemnenteed by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not reqquire it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

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